CFA LEVEL I / STUDY

另类投资
双语速学讲义

ALTERNATIVE INVESTMENTS

Concepts first. Original questions next.

61	8	6
原题 / Questions	学习单元 / Units	费用计算题 / Fee cases

先理解知识点，再用原题验证

完整保留英文题干、A/B/C选项与原题号

中文翻译 + 双语解析 + 易错点 + 公式速查

Built from the two supplied PDFs, with all 61 questions translated, grouped and checked against the supplied answers.

资料年份未注明 / Source edition not stated

编排日期 / Prepared: 2026-11-02

基于原题重组的学习单元；不冒充官方 Learning Module 编号。

Study units are editorial groupings, not claimed official module numbers.

阅读方法与资料范围

How to use this guide

本讲义把两份文件中的61道另类投资原题整理成8个教学单元。每个单元先教知识点，再完整展示相关原题、中文翻译、原答案、双语解析和易错点。题号始终沿用原题，方便回查。

This guide organizes the 61 supplied alternative-investment questions into eight study units. Each unit teaches the concepts first and then presents the complete related questions, translations, supplied answers, bilingual explanations and traps. Original question numbers are retained.

原资料 Source	使用范围 Scope
1000题(6).pdf Supplied question bank	总PDF第104-114页；章内页1-11；Q1-Q61。 PDF pages 104-114; chapter pages 1-11; Q1-Q61.
另类投资(2).pdf Supplied answer file	共61页；第n页对应Qn；全部答案已逐题匹配。 61 pages; page n corresponds to Qn; every answer has been matched.
年份与出处 Edition and provenance	原文件未明示教材年份，也未明确标注官方题或模拟题；本讲义不据文件元数据推断。 No explicit curriculum year or official-versus-mock identification is given; file metadata is not used to infer either.

学习单元名称和顺序为便于学习而编排，不声称对应某一年度官方Learning Module编号。英文原题、数字和选项不改写；仅重排换行和表格。解析依据所附答案重新撰写，并对明确笔误及概念争议作说明。

Unit names and sequence are editorial, not claimed official Learning Module numbers for a particular edition. English wording, figures and options are retained, with line breaks and tables reformatted. Explanations are rewritten from the supplied solutions, with explicit typos and conceptual issues identified.

> 记忆抓手 / Recall cue
>
> 先遮住答案作答，再读中文解析，最后用英文关键词复述。每个单元末尾的原题全部保留，既能顺着学习，也能按题号跳转复习。
> Cover the answer before attempting a question, read the Chinese explanation, and then restate the rule using the English terms. All questions are retained after their unit’s concept notes for either sequential study or question-based review.

资料校核集中在Q1、Q2、Q3、Q6、Q9、Q19、Q24和Q45；Q52另作分类确认。详细说明见附录。外部资料仅用于核对术语和时间语境，没有加入外部题目或自编题。

Context or error notes concern Q1, Q2, Q3, Q6, Q9, Q19, Q24 and Q45, with a classification confirmation for Q52. Details are in the appendix. External sources only check terminology or time context; no external or invented questions are added.

建议首轮：120分钟

Suggested first pass: 120 minutes

以下是学习时间分配建议，不是考试权重或掌握效果保证。先建立框架，再用代表原题检查理解；第二轮再完成每个单元的其余题目。

This is a suggested study allocation, not exam weights or a guarantee of mastery. Establish the framework first and check understanding with selected original questions; complete the remaining questions on a second pass.

时间 Time	任务 Task	先做这些原题 First-pass questions
15分钟 15 min	单元1：投资结构 Unit 1: investment routes	Q55、Q16、Q32、Q34
25分钟 25 min	单元2：费用计算与条款 Unit 2: fees and terms	Q61、Q24、Q19、Q27、Q58
15分钟 15 min	单元3：私募资本 Unit 3: private capital	Q8、Q11、Q36、Q50
10分钟 10 min	单元4：房地产 Unit 4: real estate	Q40、Q52、Q56
10分钟 10 min	单元5：基础设施 Unit 5: infrastructure	Q18、Q31、Q54
10分钟 10 min	单元6：自然资源 Unit 6: natural resources	Q17、Q48、Q53
20分钟 20 min	单元7：对冲基金 Unit 7: hedge funds	Q39、Q42、Q44、Q7、Q30、Q59
10分钟 10 min	单元8：数字资产 Unit 8: digital assets	Q2、Q4、Q6、Q1
5分钟 5 min	附录B：闭卷复述 Appendix B: closed-book recall	重做最不确定的原题 / Revisit uncertain items

第二轮重点：把六道计算题Q61、Q24、Q19、Q49、Q45、Q46按这一顺序独立重做；每次先写A0、A1、M、H、I，再检查题目究竟问哪一个结果。其余概念题以“为什么另外两个选项不合适”为复盘标准。

Second pass: independently redo the six fee questions in this order: Q61, Q24, Q19, Q49, Q45, Q46. Write A0, A1, M, H and I, then identify the requested result. For conceptual questions, explain why the other two options do not fit.

> 记忆抓手 / Recall cue
>
> 本题组最密集的是对冲基金策略 / 流动性 / 估值（12题）以及费用 / 绩效 / 分配（11题）。这仅描述这61题的构成，不代表CFA考试出题频率。
>
> The largest groups here are hedge fund strategy/liquidity/valuation (12 questions) and fees/performance/distribution (11). These counts describe only the supplied 61 questions and are not CFA exam frequencies.

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每个单元按“知识点 → 原题示例”排列。PDF书签可直接跳到单元或某一道原题；附录C按原题号列出本讲义页码及两份原资料页码。

Each unit follows concepts with original examples. PDF bookmarks navigate to a unit or individual question. Appendix C lists guide pages and both source-page references in original question-number order.

单元 1 另类投资特点与投资结构

Alternative Investment Features and Structures

先分清“买什么”和“怎么买”，再比较控制权、专业要求与费用。

Separate the asset or strategy from the route used to invest; then compare control, expertise and fees.

对应原题 / Original questions: Q55, Q23, Q57, Q32, Q16, Q15, Q34, Q41, Q28 | 9题 / questions

1. 另类投资的边界

1. The boundary of alternative investments

传统投资通常指股票、债券、现金的多头持仓；另类投资超出这一范围，包括私募资本、房地产、基础设施、自然资源、对冲基金及数字资产等。黄金和房地产不是本题中的传统投资，现金才是。另类既可以是有形资产，也可以是无形资产。对应 Q23、Q55、Q57。

Traditional investments are generally long-only positions in stocks, bonds and cash. Alternatives extend beyond that scope and include private capital, real estate, infrastructure, natural resources, hedge funds and digital assets. In this set, cash is traditional; gold and real estate are not. Alternatives can be tangible or intangible. See Q23, Q55 and Q57.

常见特点是流动性较低、估值更复杂、信息透明度较低、需要较多尽职调查。与传统资产的低相关性可能带来分散效果，但不是每种另类资产、每个市场时期都低相关。不要把“另类”理解为一定高收益或一定低风险。对应 Q12、Q23、Q33、Q43、Q56。

Common features include limited liquidity, complex valuation, less transparency and greater due diligence needs. Lower correlation may provide diversification, but it does not hold for every alternative or every period. Alternative does not mean guaranteed high return or low risk. See Q12, Q23, Q33, Q43 and Q56.

2. 三种投资方式：按控制权比较

2. Three routes: compare control

方式 Route	要记住的区别 Key distinction	原题 Questions
基金投资 Fund investing	把选择、执行与管理交给基金经理；直接参与要求最低，投资者控制权通常最少。 Delegate selection and management; generally the least investor expertise and control.	Q32
共同投资 / 跟投 Co-investing	通过基金投资，同时有权与基金一起直接投资某项资产；是权利，不是义务。控制权低于完全直接投资；通常比纯基金投资费用低。 Invest through a fund plus the right, not an obligation, to invest alongside it. Less control than direct investing; generally lower fees than fund-only investing.	Q15 / Q34 / Q41
直接投资 Direct investing	自行选资产、谈条件、管理和退出；灵活性、控制权及专业要求通常最高。 Select, negotiate, manage and exit directly; generally the greatest flexibility, control and expertise requirements.	Q16

Control / expertise: Fund < Co-investing < Direct

这是常见比较方向，不是所有合同的严格排序；费用也应看具体安排。

This is a typical comparison, not an absolute ranking for every contract; fees depend on the arrangement.

3. 结构不等于资产类别

3. A structure is not an asset class

Joint venture（合资企业）是合作持有和运营资产的结构。在本题组中，它与房地产直接投资对应：一方可能提供资金，另一方提供当地开发或管理能力。不能把"使用合资结构"误认成一种独立资产。对应 Q28。

A joint venture is a structure for jointly owning and operating assets. In this set it is associated with direct real estate investing, potentially combining capital with local development or management expertise. It is not a separate asset class. See Q28.

> 记忆抓手 / Recall cue
>
> 答题抓手：least expertise → fund；most flexibility → direct；right to invest alongside → co-investing。先看比较对象，再回答"更高 / 更低"。
>
> Recognition cues: least expertise → fund; most flexibility → direct; right to invest alongside → co-investing. Identify the comparison before choosing higher or lower.

原题示例与逐题解析

Original questions and worked explanations

Q55 传统投资的边界

The boundary of traditional investments

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 55页。

Which of the following is most appropriately categorized as a traditional investment?

以下哪一项最适合归类为传统投资？

A	Gold	黄金。
B	Cash	现金。
C	Real estate	房地产。

答案 / Supplied answer: B

中文解析

传统投资通常包括股票、债券和现金。黄金属于商品，房地产属于实物资产，两者在本题分类中都属于另类投资。

Explanation

Traditional investments generally include stocks, bonds and cash. Gold is a commodity and real estate is a real asset; both are classified as alternative investments in this question.

易错点 / Trap: 黄金虽然常被持有，但"常见"不等于"传统资产类别"。

An asset being widely held does not make it part of the traditional asset categories.

Q23 另类投资的定义与典型特征

Definition and typical features of alternatives

来源 / Source: 题库 PDF 107页（章内 4页）；答案 PDF 23页。

Which of the following statements is most accurate? Alternative investments:

以下哪项陈述最准确？另类投资：

A	tend to be more efficiently priced than traditional investments.	通常比传统投资的定价更有效。
B	fall outside of the definition of long-only positions in stocks, bonds, and cash.	不属于仅做多股票、债券及现金的传统投资定义范围。
C	have relatively high correlation of returns with those of traditional investments.	其收益与传统投资收益的相关性相对较高。

答案 / Supplied answer: B

中文解析

题库按投资范围及策略界定另类投资：它们不局限于股票、债券和现金的传统只做多持仓，因此选B。许多另类市场透明度和流动性较低，不能概括为比传统市场定价更有效，A不当。分散化价值常来自不同的收益来源及较低相关性，C不是一般性定义；实际相关性仍因资产和时期而异。

Explanation

The source defines alternatives as investments outside traditional long-only positions in stocks, bonds and cash, so B is correct. Many alternative markets have less transparency and liquidity, making A an unsuitable generalization. Diversification often comes from different return drivers and lower correlations, so C is not a defining feature; actual correlations vary by asset and period.

易错点 / Trap: 另类投资不是保证低相关或高收益，而是传统资产或传统持有策略范围以外的投资。

Alternatives are defined by assets or strategies outside the traditional scope, not by guaranteed low correlation or high returns.

Q57 有形与无形另类资产

Tangible and intangible alternative assets

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 57页。

Alternative investments:

另类投资：

A	include tangible assets only.	仅包括有形资产。
B	include intangible assets only.	仅包括无形资产。
C	may include both tangible and intangible assets.	可以同时包括有形和无形资产。

答案 / Supplied answer: C

中文解析

另类投资的范围很广：房地产、林地等属于有形资产；数字资产等可以属于无形资产。因此不能用是否有形来限定整个另类投资类别。

Explanation

Alternative investments span tangible assets, such as real estate and timberland, and intangible assets, such as digital assets. Tangibility therefore does not define the boundary of the entire alternative-investment category.

易错点 / Trap: 另类投资是资产与策略的广泛分类，不等于"实物资产"的同义词。

Alternative investment is a broad asset-and-strategy classification, not a synonym for real assets.

Q32 投资方式与专业能力

Investment methods and expertise

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 32页。

Which of the following methods of investing in alternative investments requires the least amount of investment expertise?

以下哪种另类投资方式对投资者所需的投资专业能力要求最低？

A	Co-investing	共同投资。
B	Fund investing	基金投资。
C	Direct investing	直接投资。

答案 / Supplied answer: B

中文解析

基金投资把具体资产的选择和管理交给专业基金经理，因此相对最少需要投资者自行具备资产层面的专业能力。共同投资需要评估具体项目，直接投资更需要自行开展尽职调查及管理。

Explanation

Fund investing delegates asset selection and management to a professional manager, requiring the least asset-level expertise from the investor. Co-investing requires assessment of individual deals, while direct investing requires more due diligence and management capability.

易错点 / Trap: “专业能力要求最低”不等于不需要评估基金经理。

The least expertise required does not mean that manager due diligence is unnecessary.

Q16 投资方式的灵活性

Flexibility of investment methods

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 16页。

Which of the following methods of investing in alternative investments provides the most flexibility to the investor?

以下哪一种另类投资方式能为投资者提供最大的灵活性？

A	Co-investing	共同投资。
B	Fund investing	基金投资。
C	Direct investing	直接投资。

答案 / Supplied answer: C

中文解析

直接投资允许投资者自行决定标的、结构、持有及退出安排，通常最灵活，因此选C。基金投资需要接受基金既定授权和经理决策；共同投资可参与特定交易，但仍受到主导基金交易安排的影响。灵活性越高，通常也要求更多专业能力和管理资源。

Explanation

Direct investing gives the investor the greatest discretion over asset selection, structuring, holding and exit decisions, so C is correct. Fund investing follows a fund mandate and manager decisions. Co-investing permits participation in selected transactions but remains influenced by the lead fund's arrangements. Greater flexibility generally requires more expertise and resources.

易错点 / Trap: 最大灵活性对应直接投资，也意味着更高的投资者能力要求。

The greatest flexibility comes with direct investing and greater demands on investor expertise.

Q15 共同投资与直接投资的控制权

Control in co-investing and direct investing

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 15页。

Compared with direct investing, co-investing in alternative investments most likely offers:

与直接投资相比，另类投资中的共同投资最可能提供：

A	reduced control over the investment selection process.	较少的投资选择过程控制权。
B	the same level of control over the investment selection process.	相同程度的投资选择过程控制权。
C	higher control over the investment selection process.	较高的投资选择过程控制权。

答案 / Supplied answer: A

中文解析

直接投资由投资者自行寻找、筛选并管理交易，控制权通常最大。共同投资者虽可选择是否参与基金经理提供的交易，但交易来源、结构及执行仍在较大程度上依赖主导方，因此相对直接投资控制权较低，选A。B及C高估了共同投资相对于直接投资的自主程度。

Explanation

A direct investor sources, selects and manages transactions and generally has the greatest control. A co-investor may decide whether to join a manager-led deal, but sourcing, structuring and execution still depend substantially on the lead party. Co-investing therefore offers less control than direct investing, making A correct; B and C overstate its relative autonomy.

易错点 / Trap: 先看比较对象：共同投资比基金投资更有参与度，但比直接投资控制权少。

Check the comparison: co-investing offers more involvement than fund investing, but less control than direct investing.

Q34 共同投资的权利

Co-investment rights

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 34页。

In co-investing, the investor invests in alternative assets indirectly through a fund but also has the:

在共同投资中，投资者通过基金间接投资另类资产，同时还拥有什么？

A	right to invest directly in the same assets alongside the fund.	与基金一同直接投资相同资产的权利。
B	obligation to invest directly in the same assets alongside the fund.	与基金一同直接投资相同资产的义务。
C	right to invest in the general partner's fund management company.	投资普通合伙人的基金管理公司的权利。

答案 / Supplied answer: A

中文解析

共同投资允许基金投资者选择在基金之外，直接参与基金所投的同一项目。这是额外投资的权利，而非必须履行的义务；投资对象是底层资产，不是基金管理公司。

Explanation

Co-investing gives a fund investor the option to invest directly alongside the fund in the same underlying deal. It is a right, not an obligation, and the investment is in the asset rather than the fund management company.

易错点 / Trap: 抓住right与obligation，以及underlying asset与management company的区别。

Distinguish a right from an obligation, and the underlying asset from the management company.

Q41 共同投资与费用

Co-investing and fees

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 41页。

Compared with fund investing in alternative investments, the co-investing method most likely has:

与通过基金投资另类资产相比，共同投资方式最可能具有怎样的管理费？

A	lower management fees.	更低。
B	the same level of management fees.	相同。
C	higher management fees.	更高。

答案 / Supplied answer: A

中文解析

共同投资的直接项目部分通常适用较低费用，或免收某些基金层面的费用，因此总体上管理费更低。投资者也通常需要自行投入更多项目分析和尽职调查资源。

Explanation

The direct deal portion of co-investing often carries lower fees or avoids some fund-level charges, so management fees are generally lower. Investors usually contribute more of their own deal-analysis and due-diligence resources.

易错点 / Trap: 费用较低不表示专业能力或尽调要求也较低。

Lower fees do not imply lower expertise or due-diligence requirements.

Q28 合资企业与房地产直接投资

Joint ventures and direct real estate

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 28页。

A joint venture is an alternative investment structure that is most likely used for.

合资企业是一种另类投资结构，最可能用于：

A	infrastructure investment.	基础设施投资。
B	private equity investment.	私募股权投资。
C	real estate direct investment.	房地产直接投资。

答案 / Supplied answer: C

中文解析

本题把合资结构与房地产直接投资联系：资金提供方可以与具备开发或物业管理经验的合作方共同拥有项目，分工投入资本及运营能力，因此选C。A及B并非绝对不能采用合资结构，但不是本题要求的最典型对应关系。

Explanation

The source associates joint ventures with direct real estate investment, where capital providers can partner with developers or property-management specialists and share ownership and responsibilities. C is correct. Infrastructure and private equity can also use joint ventures, but A and B are not the most characteristic association tested here.

易错点 / Trap: “最可能使用”不是“只有这一类能够使用”。

"Most likely used" does not mean "used exclusively by this category."

单元 2 费用、绩效与利益分配

Fees, Performance and Distribution

把六道计算题统一为一套步骤，再分清门槛收益、高水位、分配瀑布与追回条款。

Use one sequence for the six fee calculations, then distinguish hurdles, high-water marks, waterfalls and clawbacks.

对应原题 / Original questions: Q25, Q61, Q24, Q19, Q49, Q45, Q46, Q27, Q29, Q58, Q37 | 11题 / questions

1. 先识别计算基数

1. Identify the fee base first

本题组的对冲基金管理费按 AUM（管理资产规模）收取；Q25、Q37 对私募股权使用承诺资本的典型口径。承诺资本 committed capital 是承诺投入的总额；已调用 / 已催缴资本 called capital 是GP已通知投资者缴付的金额；已缴资本 paid-in capital 才是实际已缴付金额，三者不能混用。实际基金在不同阶段可以改变费用基数，应以条款为准。

Hedge fund management fees in this set are based on AUM. Q25 and Q37 use the typical private equity convention of committed capital. Committed capital is promised funding; called capital is the amount the GP has requested; paid-in capital is the amount actually contributed. These measures are not interchangeable. Actual fund agreements may change the fee base over the life of the fund.

六道计算题均要求先扣管理费再算业绩报酬。读到 based on end-of-year AUM before fees，就用“年末、扣费前”资产作为管理费基数，不能误用期初 AUM。题目如已直接给年末 AUM，不要再乘一次年度收益率。

All six calculation questions charge incentive fees net of management fees. When the management fee is based on end-of-year AUM before fees, use that year-end gross asset value, not beginning AUM. If year-end AUM is already supplied, do not apply the annual return again.

符号 Symbol	含义 Meaning	单位 Units
A0 / A1	期初 AUM / 期末扣费前 AUM Beginning AUM / ending AUM before fees	金额 Amount
g / m / p / h	毛收益率 / 管理费率 / 业绩费率 / 硬门槛收益率 Gross return / management fee rate / incentive fee rate / hard hurdle rate	小数 Decimals
M / H / I	管理费 / 门槛收益金额 / 业绩报酬 Management fee / hurdle amount / incentive fee	金额 Amount

2. 六道原题共用的计算步骤

2. The common calculation sequence

$$A1 = A0 \times (1 + g)$$

第 1 步：算年末扣费前 AUM。若题目直接给出 A1，直接使用。
Step 1: calculate ending AUM before fees, or use A1 if supplied.

$$M = m \times A1$$

第 2 步：先算管理费。本题组计算题均以年末扣费前 AUM 为基数。
Step 2: calculate the management fee using ending AUM before fees for this set.

$$H = h \times A0$$

第 3 步：把硬门槛收益率转换为金额。没有门槛时设 H = 0。
Step 3: express the hard hurdle as an amount. Set H = 0 when there is no hurdle.

$$I = p \times \max(0, A1 - A0 - M - H)$$

第 4 步：先扣管理费，再扣硬门槛；只对正的超额利润收取业绩报酬。
Step 4: deduct the management fee and hard hurdle, then charge the incentive fee only on positive excess profit.

$$\text{Total fees} = M + I;\ R_{net} = (A1 - A0 - M - I) / A0$$

第 5 步：看清题目问业绩费、总费用还是投资者净收益率。
Step 5: check whether the question asks for the incentive fee, total fees or the investor's net return.

理解提示 / Context note

公式范围：无申购赎回、按年末扣费前 AUM 收管理费、业绩费在管理费之后计算、使用硬门槛。Q19、Q45、Q46、Q49有硬门槛；Q24、Q61没有门槛。高水位和软门槛如出现，需要按条款调整，不能机械套用。
Scope: no external flows, management fees on ending gross AUM, incentives net of management fees, and a hard hurdle. Q19, Q45, Q46 and Q49 have a hard hurdle; Q24 and Q61 have none. A high-water mark or soft hurdle requires additional contract-specific treatment.

3. 用原题 Q19 跑通全过程

3. Walk through original Q19

单位为百万美元：期初 500，毛收益率 22%，管理费率 2%，业绩费率 15%，硬门槛 10%。先算 A1 = 500 × 1.22 = 610；M = 610 × 2% = 12.2；H = 500 × 10% = 50；I = (610 - 500 - 12.2 - 50) × 15% = 7.17，最接近 7.2，因此选 A。这里题目只问业绩费，不是总费用。

Amounts are in USD millions: A0 = 500, gross return = 22%, management fee = 2%, incentive fee = 15%, hard hurdle = 10%. A1 = 500 × 1.22 = 610; M = 610 × 2% = 12.2; H = 500 × 10% = 50; I = (610 - 500 - 12.2 - 50) × 15% = 7.17, closest to 7.2, so A. The question asks for the incentive fee alone, not total fees.

4. 四组条款：保护的是不同问题

4. Four terms address different problems

条款 Term	作用 Purpose	原题 Questions
硬门槛 / 软门槛 Hard / soft hurdle	硬门槛只对超出最低收益的部分收费；软门槛达到触发条件后，收费范围可能扩大到全部利润，具体以约定为准。 A hard hurdle charges only excess profit; a soft hurdle may charge all profit once triggered, subject to its terms.	Q19 / Q45 / Q46 / Q49
高水位 High-water mark	避免对已赚过、后来亏掉、再赚回的同一段业绩重复收业绩费。 Avoids charging twice for the same performance after recovery from losses.	Q27
整体基金分配 Whole-of-fund waterfall	把基金整体结果纳入分配，通常较有利于 LP；逐笔项目分配可能让 GP 较早从赢家项目收费。 Considers aggregate fund results and generally favors LPs; deal-by-deal allocation can pay GP carry earlier on winning deals.	Q29
追回条款 Clawback	在符合约定时，LP 可追回 GP 多收的部分业绩报酬；不是保证收回全部投资亏损。 Allows LPs to reclaim excess GP performance fees under agreed terms; it does not insure investment losses.	Q58

5. J 曲线：先支出，后收获

5. The J-curve: outlays before realizations

私募股权投资初期通常先发生资本调用与费用，退出收益较晚出现，因此现金流和早期表现常呈先弱后强的 J 曲线特征。基金经理可影响投资和退出时点；不能说其对分配时点完全没有裁量权。对应 Q37。

Private equity typically has early capital calls and fees, with realizations arriving later. Cash-flow patterns and early performance therefore often display a J-curve. Managers can influence investment and exit timing; they do not have zero discretion over distributions. See Q37.

> 记忆抓手 / Recall cue
>
> 最容易丢分的三处：期初还是期末；gross 还是 net of management fees；题目要 fee amount 还是 return。先写单位，再动计算器。
>
> Three common errors: beginning versus ending AUM; gross versus net of management fees; and fee amount versus return. Write the units before using the calculator.

原题示例与逐题解析

Original questions and worked explanations

Q25 对冲基金与私募股权的管理费基数

Management-fee bases for hedge funds and private equity

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 25页。

Management fees are most likely based on assets under management for:

以下哪一类基金的管理费最可能以管理资产规模（AUM）为基础计算？

A	hedge funds only.	仅对冲基金。
B	private equity funds only.	仅私募股权基金。
C	both hedge funds and private equity funds.	对冲基金和私募股权基金两者。

答案 / Supplied answer: A

中文解析

本题采用典型收费惯例：对冲基金管理费通常按AUM或资产净值计算；私募股权基金在投资期通常按承诺资本计算，因此选A。B及C忽略了这一基数差别。私募股权基金在后续阶段可能改用已投资资本等基数，实际仍须查阅合同，而不能把本题惯例当作所有基金的永久规则。

Explanation

Under the typical convention tested here, hedge fund management fees are based on AUM or NAV, whereas private equity fees during the investment period are often based on committed capital. A is therefore correct. B and C overlook this difference. Private equity fee bases may change to invested capital or another measure later in the fund life; actual contracts govern.

易错点 / Trap: 典型对照：hedge fund看AUM；PE投资期看committed capital。

Typical comparison: hedge fund fees use AUM; private equity investment-period fees use committed capital.

Q61 无门槛时的净回报

Net return without a hurdle

来源 / Source: 题库 PDF 114页（章内 11页）；答案 PDF 61页。

An analyst gathers the following information about a hedge fund:

Beginning-of-year assets under management (AUM)	$50 million
Annual return before fees	20%
Management fee (based on end-of-year AUM before fees)	2%
Incentive fee (based on return net of management fees)	20%

An investor's net return is:

分析师获得某对冲基金的信息：期初管理资产为$50 million；年度费前回报为20%；管理费为年末费前AUM的2%；业绩报酬为扣除管理费后回报的20%。投资者的净回报是多少？

A	13.60%	13.60%。
B	14.08%.	14.08%。
C	14.40%.	14.40%。

答案 / Supplied answer: B

中文解析

金额单位为百万美元。年末费前AUM=50×1.20=60；管理费M=60×2%=1.20；业绩报酬I=(60-50-1.20)×20%=1.76；总费用=2.96。净回报=(60-50-2.96)/50=14.08%。A计算业绩报酬时没有先扣管理费；C错误地以期初AUM计算管理费。

Explanation

Amounts are in USD millions. Year-end pre-fee AUM=50×1.20=60; M=60×2%=1.20; I=(60-50-1.20)×20%=1.76. Total fees=2.96. Net return=(60-50-2.96)/50=14.08%. A fails to deduct management fees before calculating the incentive fee. C incorrectly bases management fees on beginning AUM.

易错点 / Trap: 先算年末费前AUM，再算管理费，再算业绩报酬，最后除以期初AUM。

Calculate year-end pre-fee AUM, then management fees, then incentive fees, and finally divide net profit by beginning AUM.

Q24 没有门槛时的总费用

Total fees without a hurdle

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 24页。

An analyst gathers the following information about a hedge fund:

$200 million in assets under management at the beginning of year

A 2% management fee based on year-end assets under management

a 20% incentive fee calculated net of the management fee

If the fund's gross return is 25% during the year, the total fees earned by the fund manager are:

某分析师收集到一家对冲基金的以下资料：年初AUM为2亿美元；管理费为年末AUM的2%；激励费率为20%，以扣除管理费后的收益计算。若基金当年毛收益率为25%，基金经理赚取的总费用是多少？

A	$11 million	1,100万美元。
B	$14 million.	1,400万美元。
C	$15 million.	1,500万美元。

答案 / Supplied answer: B

中文解析

金额单位为百万美元。①年末费用前AUM = 200 × 1.25 = 250。②管理费 = 250 × 2% = 5。③激励费 = (250 - 200 - 5) × 20% = 9。④总费用 = 5 + 9 = 14，选B。C对应忽略“扣管理费后”而对50的毛利润收取20%激励费；A不符合题目给定的两项收费规则。 原解析把扣除的5百万美元称作激励费，实为管理费；此处已更正名称。

Explanation

Amounts are in USD millions. (1) Ending pre-fee AUM = 200 × 1.25 = 250. (2) Management fee = 250 × 2% = 5. (3) Incentive fee = (250 - 200 - 5) × 20% = 9. (4) Total fees = 5 + 9 = 14, so B is correct. C results from charging the incentive fee on the gross profit of 50 without deducting management fees. A does not follow the specified fee rules. The source calls the deducted USD 5 million an incentive fee; it is the management fee, and the label is corrected here.

易错点 / Trap: 题目问total fees，最后须把管理费与激励费相加。

For total fees, add the management fee and incentive fee at the end.

Q19 硬门槛与扣管理费后的激励费

Hard hurdle and incentive fees net of management fees

来源 / Source: 题库 PDF 107页（章内 4页）；答案 PDF 19页。

An analyst gathers the following information about a hedge fund:

Beginning-of-year assets under management (AUM)	$500 million
Annual return before fees	22%
Management fee, based on end-of-year AUM before fees	2%
Incentive fee	15%
Hard hurdle rate	10%

The incentive fee (in $ millions) based on returns net of management fees is closest to:

某分析师收集到一家对冲基金的以下资料：年初管理资产规模（AUM）为5亿美元；费用前年收益率为22%；管理费率为2%，按费用前的年末AUM计算；激励费率为15%；硬门槛收益率为10%。若激励费以扣除管理费后的收益为基础，激励费最接近多少百万美元？

A	7.2.	7.2。
B	13.0.	13.0。
C	14.7.	14.7。

答案 / Supplied answer: A

中文解析

以下金额单位均为百万美元。①年末费用前AUM = 500 × 1.22 = 610。②管理费 = 610 × 2% = 12.2。③门槛金额 = 500 × 10% = 50。④可计激励费利润 = 610 - 500 - 12.2 - 50 = 47.8。⑤激励费 = 47.8 × 15% = 7.17，最接近7.2，选A。硬门槛意味着只对超过门槛的利润收费；B及C没有正确同时处理管理费和硬门槛。 原答案解析把500 × 22%误写成610；此处已补回应有的“1 +”。

Explanation

All amounts are in USD millions. (1) Ending pre-fee AUM = 500 × 1.22 = 610. (2) Management fee = 610 × 2% = 12.2. (3) Hurdle amount = 500 × 10% = 50. (4) Eligible profit = 610 - 500 - 12.2 - 50 = 47.8. (5) Incentive fee = 47.8 × 15% = 7.17, closest to 7.2, so A is correct. A hard hurdle charges only profit above the hurdle. B and C fail to account correctly for both management fees and the hard hurdle. The source prints 500 × 22% = 610; the missing "1 +" has been restored here.

易错点 / Trap: 先算期末AUM，再扣管理费，再扣以期初资本计算的硬门槛。

Calculate ending AUM first, then deduct the management fee and the hard-hurdle amount based on starting capital.

Q49 硬门槛与净额计费

Hard hurdles and net-of-management fees

来源 / Source: 题库 PDF 112页（章内 9页）；答案 PDF 49页。

An analyst gathers the following information about a hedge fund:

Beginning-of-year assets under management (AUM)	€200 million
Management fee, based on end-of-year AUM before fees	1.5%
Incentive fee	20%
Hard hurdle rate	5%
Annual return before fees	15%

If the incentive fee is based on returns net of management fees, total fees for the year are closest to:

分析师获得某对冲基金的信息：期初管理资产为€200 million；管理费为年末费前AUM的1.5%；业绩报酬20%；硬门槛收益率5%；年度费前回报15%。若业绩报酬按扣除管理费后的回报计算，全年总费用最接近多少？

A	€6.8 million.	€6.8 million。
B	€7.5 million.	€7.5 million。
C	€8.8 million.	€8.8 million。

答案 / Supplied answer: A

中文解析

金额单位为百万欧元。年末费前AUM=200×1.15=230；管理费M=230×1.5%=3.45；硬门槛金额=200×5%=10；业绩报酬I=(230-200-3.45-10)×20%=3.31。总费用=3.45+3.31=6.76，最接近€6.8 million。B忽略管理费对业绩报酬基数的影响；C漏减硬门槛金额。

Explanation

Amounts are in EUR millions. Year-end pre-fee AUM=200×1.15=230; M=230×1.5%=3.45; hard-hurdle amount=200×5%=10. I=(230-200-3.45-10)×20%=3.31. Total fees=3.45+3.31=6.76, closest to €6.8 million. B ignores the management-fee deduction in the incentive-fee base; C omits the hurdle deduction.

易错点 / Trap: 硬门槛只对超过门槛的利润收费，且本题还须先扣管理费。

A hard hurdle charges only on profit above the hurdle, after management fees in this question.

Q45 净回报与硬门槛

Net return with a hurdle

来源 / Source: 题库 PDF 111页（章内 8页）；答案 PDF 45页。

A hedge fund has the following characteristics:

Assets under management, beginning of year	$100 million
Assets under management, end of year	S120 million
Management fee	1% of year-end assets under management
Performance fee	15% of the annual return above a 3% hurdle rate

If the performance fee is calculated net of the management fee and there were no capital contributions or withdrawals, the net annual return to the investor is closest to:

某对冲基金具有以下特征：期初管理资产（AUM）为$100 million；年末管理资产原文印作“S120 million”，答案文件按$120 million计算；管理费为年末AUM的1%；业绩报酬为超过3%门槛收益率部分的15%。若业绩报酬在扣除管理费后计算，且期间没有资本投入或提取，投资者的年度净回报最接近多少？

A	16.3%.	16.3%。
B	16.4%.	16.4%。
C	16.5%.	16.5%。

答案 / Supplied answer: B

中文解析

原题年末金额的货币符号印作S；以下按答案文件明确使用的$120 million计算。金额单位均为百万美元。管理费M=120×1%=1.20；门槛金额=100×3%=3.00；可提取业绩报酬的利润=120-100-1.20-3.00=15.80；业绩报酬I=15.80×15%=2.37。净回报=(120-100-1.20-2.37)/100=16.43%，最接近B的16.4%。A、C均不等于正确计费顺序的结果。

Explanation

The source prints S before the year-end amount; the calculation below follows the answer file’s explicit $120 million. All amounts are in USD millions. Management fee M=120×1%=1.20. The hurdle amount is 100×3%=3.00. Eligible profit=120-100-1.20-3.00=15.80; incentive fee I=15.80×15%=2.37. Net return=(120-100-1.20-2.37)/100=16.43%, closest to B, 16.4%. A and C do not match the correct fee sequence.

易错点 / Trap: 管理费用年末AUM；门槛金额和投资者回报分母用期初AUM。

Use year-end AUM for the management fee, but beginning AUM for the hurdle amount and return denominator.

Q46 总费用与业绩报酬

Total fees versus incentive fees

来源 / Source: 题库 PDF 111页（章内 8页）；答案 PDF 46页。

An analyst collects the following information about a hedge fund:

Assets under management	$1.5 billion, beginning of year
Annual return	20%
Management fee	2%, based on year-end valuation
Incentive fee	20%, calculated net of management fees

If the incentive fee is calculated on returns in excess of a 6% hurdle rate, total annual fees earned by the fund manager are closest to:

分析师收集到某对冲基金的信息：期初管理资产为$1.5 billion；年度回报20%；管理费为年末估值的2%；业绩报酬为20%，按扣除管理费后的回报计算。若业绩报酬只针对超过6%门槛收益率的回报收取，基金管理人年度总费用最接近多少？

A	$34,800,000.	$34,800,000。
B	$70,800,000.	$70,800,000。
C	$78,000,000.	$78,000,000。

答案 / Supplied answer: B

中文解析

金额单位为百万美元。年末费前AUM=1,500×1.20=1,800；管理费M=1,800×2%=36；门槛金额=1,500×6%=90；业绩报酬I=(1,800-1,500-36-90)×20%=34.8。总费用=36+34.8=70.8，即$70,800,000。A只计算了业绩报酬；C没有先扣管理费再计算业绩报酬。

Explanation

Amounts are in USD millions. Year-end pre-fee AUM=1,500×1.20=1,800. M=1,800×2%=36; hurdle amount=1,500×6%=90. I=(1,800-1,500-36-90)×20%=34.8. Total fees=36+34.8=70.8, or $70,800,000. A gives only the incentive fee. C fails to deduct management fees before calculating the incentive fee.

易错点 / Trap: 题目问total fees，最后必须把管理费和业绩报酬相加。

When the question asks for total fees, add the management fee and incentive fee.

Q27 高水位线

High-water mark

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 27页。

A feature that protects hedge fund clients from paying twice for the same performance is most likely a:

以下哪一项机制最可能防止对冲基金客户就同一段业绩支付两次费用？

A	Discount.	折扣。
B	hurdle rate.	门槛收益率。
C	high-water mark.	高水位线。

答案 / Supplied answer: C

中文解析

高水位线要求基金先恢复到先前用于计算业绩费的高点，超过该水平的新收益才再计业绩费，从而避免亏损恢复阶段重复收费，选C。门槛收益率是开始收费的最低收益要求；折扣只是降低费率，均不能直接解决重复收费问题。具体高水位线口径由合同规定。

Explanation

A high-water mark prevents another performance fee on recovery to a previously charged high point; only new gains above the relevant mark become eligible again. C is correct. A hurdle specifies a minimum return requirement, while a discount reduces a fee rate. Neither directly prevents charging twice for recovered performance. The contract defines the precise mark.

易错点 / Trap: 高水位线看"过去是否已收过费"；门槛看"本期是否达到最低回报"。

A high-water mark asks whether gains were charged before; a hurdle asks whether the required return has been reached.

Q29 整只基金分配瀑布

Whole-of-fund waterfall

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 29页。

Regarding distribution methods in alternative investments, which of the following is most advantageous to the limited partners? A(n):

关于另类投资的收益分配方式，以下哪一种最有利于有限合伙人？

A	American waterfall.	美式分配瀑布。
B	deal-by-deal waterfall.	逐笔交易分配瀑布。
C	whole-of-fund waterfall.	整只基金分配瀑布。

答案 / Supplied answer: C

中文解析

整只基金分配瀑布按基金整体结果确定分成，通常先使LP收回所要求的资本及适用的优先回报，再向GP支付业绩分成，因此对LP更有利，选C。美式瀑布通常按逐笔交易结算，与B所指方法相近，可能较早对盈利交易支付分成，即使其他交易后来亏损。

Explanation

A whole-of-fund waterfall determines carry using aggregate fund results, typically returning the required capital and applicable preferred return to LPs before paying the GP's share. This is more favorable to LPs, so C is correct. An American waterfall generally operates deal by deal, similar to B, and may pay carry on early winners before later losses become known.

易错点 / Trap: Whole-of-fund先综合盈亏，较少出现GP先拿分成、LP后承担其他项目亏损的时点问题。

Whole-of-fund treatment aggregates outcomes before carry, reducing the timing mismatch between early GP payouts and later portfolio losses.

Q58 业绩报酬回拨

Clawback provisions

来源 / Source: 题库 PDF 114页（章内 11页）；答案 PDF 58页。

In alternative investments, a clawback provision represents the right of:

在另类投资中，回拨条款（clawback provision）代表谁拥有什么权利？

A	limited partners to reclaim performance losses.	有限合伙人追回投资业绩损失的权利。
B	the general partner to reclaim part of limited partners' distributions.	普通合伙人追回有限合伙人部分分配款的权利。
C	limited partners to reclaim part of the general partner's performance fee.	有限合伙人追回普通合伙人部分业绩报酬的权利。

答案 / Supplied answer: C

中文解析

若普通合伙人先前获得的业绩报酬高于其按最终约定业绩应得的金额，回拨条款允许有限合伙人追回多付部分。它调整的是管理人已获得的业绩报酬，并不承诺补偿所有投资损失，也不是普通合伙人追回投资者分配款。

Explanation

If the general partner previously received more performance fees than it is entitled to under the final agreed performance calculation, a clawback lets limited partners recover the excess. It corrects overpaid performance compensation; it does not insure investment losses or give the general partner the right described in B.

易错点 / Trap: 回拨针对多付的业绩报酬，不是本金保本或损失赔偿。

A clawback recovers excess performance compensation; it is not principal protection or loss insurance.

Q37 私募股权的J曲线

The private equity J-curve

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 37页。

Which of the following statements about private equity performance evaluation is most accurate?

以下有关私募股权绩效评估的说法哪一项最准确？

A	Private equity fund management fees are based on capital called.	私募股权基金管理费以已调用资本为基础。
B	Cash flows are frequently described in terms of the J-curve effect.	现金流经常用J曲线效应来描述。
C	Managers have no discretion on the timing of the distribution of proceeds.	管理人对收益分配的时间没有自主决定权。

答案 / Supplied answer: B

中文解析

私募股权基金早期通常有资本投入、费用及尚未实现的回报，后期随着投资成熟和退出才出现较多分配，形成先低后高的J曲线特征。题库口径下，管理费通常以承诺资本为基础，尤其在投资期内；管理人也能影响投资退出和分配的时间。

Explanation

Private equity funds typically have early capital outflows and fees, followed by distributions as investments mature and exit, producing a J-curve pattern. Under the question bank's convention, management fees are generally based on committed capital, especially during the investment period. Managers can also influence exit and distribution timing.

易错点 / Trap: 区分承诺资本与已调用资本；实际收费基础仍以基金条款和阶段为准。

Separate committed from called capital; actual fee bases depend on fund terms and the fund's stage.

单元 3 私募股权与私募债务

Private Equity and Private Debt

按企业阶段和融资工具区分策略，再辨认退出方式、偿付顺序和风险。

Classify strategies by company stage and financing instrument, then identify exits, seniority and risk.

对应原题 / Original questions: Q8, Q11, Q38, Q10, Q9, Q36, Q50, Q60 | 8题 / questions

1. 私募股权：看企业在哪一阶段

1. Private equity: identify the company stage

类别 Category	核心识别 Recognition cue	原题 Questions
风险投资 Venture capital	为早期企业提供股权资本，是 private equity 的子类别。 Equity funding for early-stage businesses; a subcategory of private equity.	Q8
成长资本 Growth capital	为已有业务的企业扩张提供资金；常见形式是少数股权投资。 Finances expansion of established businesses; often minority equity investing.	Q11
首次公开募股 IPO	把原本非上市的投资组合企业带到公开市场，是主要退出途径之一。 Takes a private portfolio company public and is a major exit route.	Q38
SPAC 退出 Exit through a SPAC	本题强调交易结构灵活；可协商估值不等于 floating valuation，SPAC 股东仍可能有赎回权。 The set emphasizes transaction flexibility. A negotiated valuation is not a floating valuation, and SPAC shareholders may retain redemption rights.	Q10

2. 私募债务：看借款人和索偿权

2. Private debt: borrower and claim

类型 Type	定义与风险抓手 Definition and risk cue	原题 Questions
风险债务 Venture debt	向早期企业提供债务资金，即使企业现金流仍很少。 Debt funding for early-stage companies even when their cash generation is limited.	Q50
困境债务 Distressed debt	购买陷入财务困难的成熟企业债务；与venture debt的企业阶段不同。 Debt of mature companies in financial difficulty; a different stage from venture debt.	Q9 / Q50
夹层债务 Mezzanine debt	通常位于优先债务之后、股权之前；可能带认股权证或转股特征，以分享股权上涨。 Generally junior to senior debt but senior to equity; may include warrants or conversion features.	Q36
单一分层债务 Unitranche debt	把优先与次级债务特征合并在一项融资中；不能因为只有一层就视为无风险。 Combines senior and subordinated features in one financing; a single tranche is not risk-free.	Q60
基础设施债务 Infrastructure debt	本题按较稳定的项目现金流理解，在Q60给定三类中风险最低；仍须看项目、结构和阶段。 Under the set's stable-cash-flow convention it is the lowest risk of Q60's three choices; project and structure still matter.	Q60

Typical claim priority: Senior debt > Mezzanine debt > Equity

这里的 > 表示偿付优先级较高，不是预期收益较高。
Here > means higher repayment priority, not higher expected return.

3. Q9 的术语口径必须单独记

3. Q9 uses a special terminology convention

理解提示 / Context note

Q9 原答案为 A：a loan that is itself levered。原解析从贷款基金先借钱、再提供贷款的口径解释。常见市场定义则把 leveraged loan 指向本身负债较高和 / 或信用评级较低的借款人。两种口径不能互相替代；本讲义保留原答案 A，不把它升级为通用定义。原题 B 在答案中归为困境债务，C 归为夹层债务。[S1]
The supplied answer to Q9 is A: a loan that is itself levered. Its explanation focuses on a lending fund borrowing to finance loans. Common market usage instead describes loans to highly indebted and/or low-rated borrowers. These conventions are not interchangeable. This guide preserves A as the supplied answer without treating it as a universal definition. The supplied solution labels B distressed debt and C mezzanine debt. [S1]

记忆抓手 / Recall cue

答题先问两个问题：这是 equity 还是 debt？企业是 early-stage、growth 还是 distressed？再看索偿顺序与退出渠道。
Ask first: equity or debt? Then: early-stage, growth or distressed? Finally consider claim priority and exit route.

原题示例与逐题解析

Original questions and worked explanations

Q8 私募股权的分类

Private equity categories

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 8页。

Venture capital is best classified as a sub-category of:

风险投资最适合被归类为以下哪一类的子类别？

A	real estate.	房地产。
B	hedge funds.	对冲基金。
C	private equity.	私募股权。

答案 / Supplied answer: C

中文解析

风险投资向初创或早期企业提供股权资本，是私募股权的一种，因此选C。房地产按实物资产或相关权益分类；对冲基金则是另一类采用多种投资策略的基金安排。不要因为三者都属于另类投资，就把它们视为同一级别的子类别。

Explanation

Venture capital supplies equity capital to start-ups or early-stage companies and is a subcategory of private equity, making C correct. Real estate concerns property assets or related interests. Hedge funds are a separate category using a range of strategies. Sharing the alternative-investment label does not make these categories interchangeable.

易错点 / Trap: 层级记忆：另类投资 → 私募股权 → 风险投资。

Remember the hierarchy: alternative investments → private equity → venture capital.

Q11 成长资本

Growth capital

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 11页。

With respect to private equity, the growth capital strategy is also known as:

在私募股权领域，成长资本策略也称为：

A	venture capital.	风险投资。
B	recapitalization.	资本重组。
C	minority equity investing.	少数股权投资。

答案 / Supplied answer: C

中文解析

成长资本通常向已有业务、需要资金扩张的企业投资，并常取得少数股权，因此又称少数股权投资，选C。风险投资更侧重初创或早期阶段；资本重组是改变企业资本结构或权益安排，与成长资本不是同义词。

Explanation

Growth capital commonly finances expansion at an established business and often takes a minority equity stake, so C is correct. Venture capital focuses more on start-ups or early-stage businesses. Recapitalization changes a company's capital structure or ownership financing and is not synonymous with growth capital.

易错点 / Trap: Growth capital：帮助已有业务扩张，通常不以取得控制权为必要条件。

Growth capital finances expansion of an existing business and does not necessarily require control.

Q38 私募股权退出

Private equity exits

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 38页。

Which of the following is most likely a primary exit strategy for a company held by a private equity fund's portfolio?

以下哪一项最可能是私募股权基金所持被投公司的主要退出策略？

A	IPO	首次公开发行（IPO）。
B	Management buy-in	外部管理层收购。
C	Management buyout	原有管理层收购。

答案 / Supplied answer: A

中文解析

IPO使被投公司进入公开市场，为私募股权投资退出或逐步变现创造途径。管理层买入和管理层收购在本题分类中属于收购安排，不是三个选项中最典型的退出方式。

Explanation

An IPO brings the portfolio company to public markets and creates a route to exit or monetize the investment over time. Management buy-ins and buyouts are acquisition arrangements in this question's classification, rather than the most typical exit route among these choices.

易错点 / Trap: IPO不一定意味着上市当天即可全部卖出；本题考的是退出方式分类。

An IPO does not necessarily permit an immediate full sale; the question tests exit-route classification.

Q10 通过SPAC退出

Exit through a SPAC

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 10页。

From the perspective of a private equity firm, an advantage of exiting a portfolio company through a special purpose acquisition company (SPAC)most likely include:

从私募股权公司的角度看，通过特殊目的收购公司（SPAC）退出一家被投资公司的优点，最可能包括：

A	floating valuation.	浮动估值。
B	flexibility of the transaction structure.	交易结构具有灵活性。
C	lower deal risk due to restrictions on redemption s.	由于赎回受到限制，交易风险较低。

答案 / Supplied answer: B

中文解析

通过与SPAC协商合并，交易各方可以协商交易结构，因而B是题目所指的优点。A的浮动估值不是所述核心优势。C把赎回安排的影响说反了：SPAC投资者的赎回权可能使交易可用资金减少，增加融资或成交不确定性，不能据此直接认定交易风险较低。

Explanation

A negotiated merger with a SPAC allows flexibility in structuring the transaction, so B is correct. A floating valuation is not the advantage identified here. C mischaracterizes redemption arrangements: SPAC investor redemptions can reduce cash available to the transaction and create financing or completion uncertainty rather than automatically reducing deal risk.

易错点 / Trap: 记住"可协商的交易结构"；不要把SPAC赎回权误记为一定降低交易风险。

Remember negotiable transaction structure; SPAC redemption rights do not automatically reduce deal risk.

Q9 杠杆贷款：原答案与市场术语口径不一致

Leveraged loans: source and market terminology differ

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 9页。

A leveraged loan is best defined as a loan:

杠杆贷款最适合被定义为以下哪一种贷款？

A	that is itself levered.	贷款本身加了杠杆。
B	to mature companies in financial difficulty.	提供给陷入财务困难的成熟企业的贷款。
C	that comes with warrants or conversion rights.	附有认股权证或转换权的贷款。

答案 / Supplied answer: A [术语争议 / Definition issue]

中文解析

按原答案选A。原解析采用贷款提供方的杠杆口径：私人债务投资机构先借入资金，再向企业放贷，因此称贷款本身带杠杆；B被原解析归为困境债务，C归为夹层债务。概念核校：常见市场定义侧重借款人负债较高和 / 或信用评级较低，并不要求放贷基金自身借款；原题口径与这一常用定义不一致，应保留原答案并标记争议，不能把A背成通用定义。

Explanation

Follow A as the supplied answer. The explanation uses leverage at the lender level: a private debt investor borrows money and then lends to a company. It assigns B to distressed debt and C to mezzanine debt. Terminology check: the usual market definition focuses on borrowers with high debt and/or low credit ratings; it does not require the lending fund itself to borrow. The source uses a different definition, so retain its answer with an explicit qualification rather than memorize A as a general definition.

易错点 / Trap: 原答案A与常见市场术语存在口径冲突；分别记住“本题答案”与“通用定义”，不要混为一谈。

The supplied A conflicts with common market terminology. Keep the source answer separate from the general market definition.

Q36 夹层债与资本结构

Mezzanine debt and the capital structure

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 36页。

Which of the following statements about private debt is most accurate? Mezzanine debt:

以下有关私募债务的说法哪一项最准确？夹层债务：

A	is subordinated to equity in a borrower's capital structure.	在借款人的资本结构中，其偿付顺序劣后于股权。
B	is less risky than senior secured debt issued by the same company.	比同一公司发行的优先有抵押债务风险更低。
C	may include additional features such as warrants to provide equity participation to lenders.	可能附带认股权证等额外条款，使贷款人参与股权收益。

答案 / Supplied answer: C

中文解析

夹层债通常位于优先债务之后、股权之前，风险高于优先有抵押债务，因此A、B都颠倒了顺序。附带认股权证或转股安排，可以让贷款人在承担较高风险的同时分享股权上涨收益。

Explanation

Mezzanine debt generally ranks below senior debt but above equity and is riskier than senior secured debt. A and B reverse these relationships. Warrants or conversion features can give lenders equity upside in compensation for higher risk.

易错点 / Trap: 偿付优先级：优先债务在前，夹层债居中，股权在后。

Priority runs from senior debt to mezzanine debt to equity.

Q50 创业债务

Venture debt

来源 / Source: 题库 PDF 112页（章内 9页）；答案 PDF 50页。

In the private debt market, venture debt:

在私募债务市场中，创业债务：

A	entails buying the debt of mature companies in financial difficulty.	涉及购买陷入财务困境的成熟公司的债务。
B	provides capital to early-stage companies that may be generating little cash flow.	向可能只产生很少现金流的早期企业提供资本。
C	entails buying the debt of mature companies in financial difficulty and provides capital to early-stage companies that may be generating little cash flow.	既涉及购买陷入财务困境的成熟公司的债务，也向可能只产生很少现金流的早期企业提供资本。

答案 / Supplied answer: B

中文解析

创业债务服务于早期企业，包括尚未产生大量经营现金流的公司。购买陷入财务困境的成熟公司的债务属于困境债务投资，不能与创业债务混同，因此A和同时包含A的C都不正确。

Explanation

Venture debt finances early-stage companies, including those with limited operating cash flow. Buying the debt of mature companies in financial difficulty is distressed-debt investing. A is therefore incorrect, and C incorrectly combines two different categories.

易错点 / Trap: 按企业阶段区分：早期企业是venture debt；成熟企业财务困境是distressed debt。

Use the company stage: early-stage financing is venture debt; mature-company financial distress points to distressed debt.

Q60 私募债务的相对风险

Relative risk of private debt

来源 / Source: 题库 PDF 114页（章内 11页）；答案 PDF 60页。

All else being equal, which of the following types of private debt is most likely to have the lowest level of risk?

其他条件相同，以下哪种私募债务最可能具有最低风险？

A	Mezzanine debt	夹层债务。
B	Unitranche debt	单一分层融资债务（unitranche debt）。
C	Infrastructure debt	基础设施债务。

答案 / Supplied answer: C

中文解析

在题库的典型比较中，基础设施债务依托长期资产及较可预测的项目现金流，风险相对最低。夹层债务的偿付顺序较低；单一分层债务将优先和次级融资特征结合，其风险通常介于相关优先与次级融资之间。

Explanation

In the source's typical comparison, infrastructure debt benefits from long-lived assets and relatively predictable project cash flows, making C the lowest-risk choice. Mezzanine debt is subordinated. Unitranche debt combines senior and subordinated financing characteristics, generally with intermediate risk.

易错点 / Trap: 这是"其他条件相同"的典型排序；项目建设阶段、杠杆和合同质量仍会改变实际风险。

This is an all-else-equal ranking; project stage, leverage and contract quality can change actual risk.

单元 4 房地产投资

Real Estate Investing

抓住异质性、交易成本、管理负担，以及 core 与 value-add 的差别。

Focus on heterogeneity, transaction costs, management burden, and the difference between core and value-add.

对应原题 / Original questions: Q43, Q56, Q35, Q40, Q52 | 5题 / questions

1. 房地产不是标准化证券

1. Property is not a standardized security

房地产具有异质性 heterogeneous：位置、物业质量、租约和用途各不相同。直接购买物业的交易成本高，资产不可轻易分割，小投资者难以通过整栋持有建立充分分散的组合。私人房地产指数通常是衡量工具，不能直接当成一篮子可交易资产来买。对应 Q43、Q56。

Real estate is heterogeneous: location, quality, leases and use vary. Direct property purchases have high transaction costs and are difficult to divide, making diversification through whole-property ownership hard for small investors. Private property indexes are generally benchmarks rather than directly investable baskets. See Q43 and Q56.

直接投资可以带来较高控制权，但也要求管理物业、租户和维护事项。Q35问的是缺点，因此答案是管理所需时间，不是“缺乏控制权”；税务待遇要看地区，不能一概说不利。

Direct investment provides control but requires time to manage the property, tenants and maintenance. Q35 asks for a disadvantage: management time, not lack of control. Tax treatment is jurisdiction-dependent rather than universally unfavorable.

2. 从稳定收租到改造增值

2. From stable income to operational improvement

策略 Strategy	辨别方式 How to identify it	原题 Questions
核心型 Core	成熟、优质、区位好的物业，现金流较稳定。Q40选择主要城市的高质量写字楼。 Established, high-quality property in a strong location with relatively stable cash flow. Q40 selects a high-quality office in a major urban center.	Q40
核心增益型 Core-plus	通常接近核心型，但承受稍多租赁、物业或融资风险；不等于所有“轻度改造”措辞。 Generally close to core with somewhat more leasing, property or financing risk; not every mention of modest work implies core-plus.	Q52对比 Q52 contrast
增值型 Value-add	通过升级、填补空置、重新定位等改善经营，再取得更高收益；Q52是本题组的代表。 Improves operations through upgrades, vacancy leasing and repositioning to earn higher returns. Q52 is the set’s example.	Q52
机会型 Opportunistic	通常涉及更大幅度开发、重建或困境处置，执行及资本增值依赖更强。 Generally involves more extensive development, redevelopment or distressed situations, with greater execution and appreciation dependence.	Q52对比 Q52 contrast

记忆抓手 / Recall cue

Q52必须整句判断：modest redevelopment + leasing vacant space + repositioning → 原答案 B，Value-add。不要仅因为看见 modest 就改选 Core-plus。
Read Q52 as a whole: modest redevelopment + leasing vacant space + repositioning → supplied answer B, Value-add. Do not choose Core-plus based on modest alone.

原题示例与逐题解析

Original questions and worked explanations

Q43 房地产的异质性

Heterogeneity of real estate

来源 / Source: 题库 PDF 111页（章内 8页）；答案 PDF 43页。

Which of the following statements about real estate assets is most accurate?

以下有关房地产资产的说法哪一项最准确？

A	Real estate assets are heterogeneous	房地产资产具有异质性。
B	Commercial property represents the majority of real estate assets by value	商业物业按价值计算占房地产资产的大多数。
C	Private real estate has historically had high correlations with other asset classes	私募房地产与其他资产类别历来具有高相关性。

答案 / Supplied answer: A

中文解析

房地产在位置、用途、质量、租约及运营情况上各不相同，因此具有异质性。按资料所用市场比较，住宅房地产而非商业物业占整体价值的较大部分；题库强调的另一潜在好处是历史上与其他资产类别的低相关性，因此B、C不正确。

Explanation

Real estate varies by location, use, quality, leases and operations, so assets are heterogeneous. In the source's market comparison, residential rather than commercial property represents the larger share of total value. The source also emphasizes historically low correlations as a potential benefit, making B and C incorrect.

易错点 / Trap: 报告中的低相关性可能受估值平滑影响；不能理解为风险必然很低。
Reported low correlations may be affected by appraisal smoothing and do not necessarily imply low risk.

Q56 私募房地产市场特征

Private real estate market characteristics

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 56页。

Which of the following is most likely a characteristic of private real estate markets?

以下哪一项最可能是私募房地产市场的特征？

A	Transaction costs are high	交易成本高。
B	Private market indexes are investable	私募市场指数可直接投资。
C	It is easy for small investors to establish a diversified portfolio of wholly owned properties	小型投资者容易建立由完全自有物业构成的分散化组合。

答案 / Supplied answer: A

中文解析

私人房地产交易通常涉及经纪、法律、尽调和其他成本，交易成本较高。私募房地产指数通常是衡量市场表现的指标，无法像买入一组公开证券那样直接复制；物业单位投资额较大，也限制小投资者分散持有全资物业。

Explanation

Private property transactions involve brokerage, legal, due-diligence and other costs. Private real estate indexes generally measure performance rather than provide a directly replicable portfolio of traded assets. Large individual property commitments also make diversification difficult for small investors.

易错点 / Trap: 不要把可投资的上市REIT指数与私募房地产估值指数混为一谈。

Do not confuse investable listed REIT indexes with private real estate appraisal indexes.

Q35 直接房地产投资的负担

Burden of direct real estate ownership

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 35页。

A disadvantage of direct real estate investing is:

直接房地产投资的一项缺点是什么？

A	a lack of control.	缺乏控制权。
B	unfavorable tax rules.	不利的税收规则。
C	the time required to manage the property.	管理物业所需的时间。

答案 / Supplied answer: C

中文解析

直接持有物业需要投入时间处理租赁、维修、运营和管理。直接投资通常给予较高控制权，所以A不合适；税务待遇取决于地区和结构，不能把"不利税制"作为普遍缺点。

Explanation

Direct ownership requires time for leasing, maintenance, operations and management. It generally provides substantial control, so A is inappropriate. Tax treatment depends on the jurisdiction and structure; unfavorable tax rules are not a universal disadvantage.

易错点 / Trap: 直接投资的控制权较高，但管理工作也更多。

Direct ownership brings greater control and greater management demands.

Q40 核心型房地产

Core real estate

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 40页。

Which of the following is best categorized as core real estate?

以下哪项最适合归类为核心型房地产？

A	A high-quality office building in a rural area	农村地区的高品质写字楼。
B	A low-quality office building in a major urban center	主要城市中心的低品质写字楼。
C	A high-quality office building in a major urban center	主要城市中心的高品质写字楼。

答案 / Supplied answer: C

中文解析

核心型房地产偏向优质物业、成熟地段和较稳定的现金流。三个选项中，C同时具备主要城市中心和高品质两项特征；A缺乏核心区位，B缺乏高品质物业。

Explanation

Core real estate emphasizes high-quality properties in established locations with relatively stable cash flows. C combines prime urban location and high quality. A lacks the core location, while B lacks the high-quality property.

易错点 / Trap: 核心型看“优质资产 + 优质区位”，不能只看其中一个。

Core classification combines property quality with location quality.

Q52 增值型房地产策略

Value-add real estate

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 52页。

Which of the following real estate investing strategies is most likely to focus on modest redevelopment or upgrades, the leasing of vacant space, and the repositioning of underlying properties to earn a higher return?

以下哪类房地产投资策略最可能着重于适度重新开发或升级、出租空置空间，以及重新定位底层物业，以取得更高回报？

A	Core-plus	核心增益型。
B	Value-add	增值型。
C	Opportunistic	机会型。

答案 / Supplied answer: B

中文解析

答案文件第52页将适度改造、出租空置空间和物业重新定位归为增值型策略：通过主动改善物业提高收入或价值。核心增益型承担略高于核心型的风险；机会型通常涉及开发、大规模重建、用途变更或更高空置风险，因此不如B贴合题意。

Explanation

Answer-file page 52 classifies modest upgrades, leasing vacant space and repositioning as value-add: active improvements seek to raise income or value. Core-plus accepts somewhat more risk than core. Opportunistic strategies usually involve development, major redevelopment, repurposing or much greater vacancy risk, making B the closest fit.

易错点 / Trap: 不要只凭modest一词选核心增益型；本题要看改造、出租和重新定位的完整组合。

Do not select core-plus solely because the word modest appears; consider the full combination of upgrades, leasing and repositioning.

单元 5 基础设施投资

Infrastructure Investing

按用途和开发阶段分类，再看现金流来源与风险如何通过合同缓解。

Classify by use and development stage, then connect return sources with contractual risk mitigation.

对应原题 / Original questions: Q22, Q18, Q31, Q33, Q54 | 5题 / questions

1. 用途：经济基础设施与社会基础设施

1. Use: economic versus social infrastructure

机场、通信塔等通常归为经济基础设施 economic infrastructure；学校、医院和惩教设施等服务社会公共需求，属于社会基础设施 social infrastructure。Q22中的 correctional facilities 是惩教设施。

Airports and telecommunications towers generally fall under economic infrastructure. Schools, hospitals and correctional facilities serve social needs and are social infrastructure. In Q22, correctional facilities are the social asset.

2. 阶段：新建项目与现有项目

2. Stage: new versus existing projects

阶段 Stage	特点与主要风险 Features and risk	原题 Questions
绿地项目 Greenfield	从新建开始，缺乏自身运营历史，承担开发和建设风险；常与开发商或战略投资者合作，较依赖未来升值。 New construction with limited operating history and development/construction risk; often alongside developers or strategic investors and more dependent on future appreciation.	Q18 / Q31
棕地项目 Brownfield	已有资产及运营记录，通常较能依赖现有收益；若主要靠资本增值，也可能承担较高风险。 Existing assets with operating history, often more dependent on current income; appreciation-led brownfield projects may still carry material risk.	Q31

Q31的比较必须同时看两个维度：开发阶段 + 回报来源。Greenfield 且主要依赖 capital appreciation（资本增值），相较题目中的两种 brownfield 方案风险最高。不能只看到“基础设施”就判低风险。

Q31 combines two dimensions: development stage and return source. Greenfield investments relying mainly on capital appreciation carry the highest risk among its three choices. Infrastructure is not automatically low risk.

3. 组合用途与 take-or-pay

3. Portfolio role and take-or-pay

基础设施可能兼有抗通胀和与现有组合低相关的优点，例如部分收入可随价格水平调整。但这取决于收费机制、合同和经济环境，并非收益保证。Q33同时选择这两项。

Infrastructure may provide both inflation protection and low correlation with existing portfolio assets, for example where revenues adjust with prices. These benefits depend on contracts, pricing and the economic environment. Q33 selects both.

Take-or-pay（照付不议）约定：买方即使没有实际取用约定数量，也须按合同付款。它主要缓解卖方的需求风险 demand risk；不会直接替你消除施工延误、设备故障或交易对手违约。对应 Q54。

Under take-or-pay, the buyer must pay under the contract even without taking the agreed volume. It primarily mitigates the seller's demand risk, not construction delays, equipment failure or counterparty default. See Q54.

记忆抓手 / Recall cue

关键词：new project → greenfield；existing operations → brownfield；pay despite low usage → demand risk protection。
Cues: new project → greenfield; existing operations → brownfield; pay despite low usage → demand-risk protection.

原题示例与逐题解析

Original questions and worked explanations

Q22 社会基础设施与经济基础设施

Social and economic infrastructure

来源 / Source: 题库 PDF 107页（章内 4页）；答案 PDF 22页。

Which of the following are best categorized as social infrastructure assets?

以下哪一项最适合被归类为社会基础设施资产？

A	Airports	机场。
B	Correctional facilities	惩教设施。
C	Telecommunication towers	电信塔。

答案 / Supplied answer: B

中文解析

社会基础设施服务于教育、医疗、司法等公共社会服务；惩教设施属于此类，因此选B。机场属于交通基础设施；电信塔属于通信基础设施，两者通常归为经济基础设施。

Explanation

Social infrastructure supports public social services such as education, health and justice. Correctional facilities fit this category, so B is correct. Airports are transportation infrastructure and telecommunications towers are communications infrastructure, both generally categorized as economic infrastructure.

易错点 / Trap: 社会服务看社会基础设施；运输、通信等经济网络看经济基础设施。
Social services indicate social infrastructure; transport and communications networks indicate economic infrastructure.

Q18 绿地基础设施投资

Greenfield infrastructure investing

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 18页。

Investors in greenfield infrastructure projects typically:

绿地基础设施项目的投资者通常：

A	rely on the assets' financial and operating history.	依赖资产已有的财务及经营历史。
B	invest alongside strategic investors or developers.	与战略投资者或开发商共同投资。
C	have lower development risk than investors in brownfield projects.	承受的开发风险低于棕地项目投资者。

答案 / Supplied answer: B

中文解析

绿地项目处于新建或开发阶段，投资者通常与掌握开发或运营能力的战略投资者、开发商合作，因此选B。此类项目尚缺乏自身成熟经营记录，A不适合。与已建成或运营中的棕地项目相比，绿地项目通常承担更高的建设和开发风险，C方向相反。

Explanation

Greenfield assets are newly developed or under construction. Investors commonly partner with strategic investors or developers that contribute development or operating expertise, so B is correct. Such projects lack an established operating history, making A unsuitable. Their construction and development risk is generally higher than for existing brownfield assets, contrary to C.

易错点 / Trap: Greenfield：未成熟运营、开发风险高、常与开发商合作。

Greenfield: no mature operating record, higher development risk, and frequent developer partnerships.

Q31 基础设施的风险排序

Infrastructure risk ranking

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 31页。

Which of the following infrastructure investments most likely have the highest risk?

以下哪类基础设施投资最可能具有最高风险？

A	Brownfield investments with the majority of their return from current yield.	回报主要来自当期收益的棕地投资。
B	Brownfield investments with the majority of their return from capital appreciation.	回报主要来自资本增值的棕地投资。
C	Greenfield investments with the majority of their return from capital appreciation.	回报主要来自资本增值的绿地投资。

答案 / Supplied answer: C

中文解析

绿地项目尚待开发建设，面临建设、完工、需求和运营等不确定性；若回报又主要依赖资本增值，未来价值更难确定，因此C最高。棕地项目已有资产及运营记录，A偏重当期现金收益，B偏重增值，但两者均少了全新项目的部分开发风险。

Explanation

Greenfield projects face development, completion, demand and operating uncertainty. Reliance on capital appreciation adds uncertainty about future value, making C the highest-risk combination. Brownfield assets already exist and have operating records; A emphasizes current income, while B emphasizes appreciation.

易错点 / Trap: 把两个维度一起看：绿地或棕地，以及当期收益或资本增值。

Combine both dimensions: greenfield versus brownfield, and current income versus capital appreciation.

Q33 基础设施的组合作用

Infrastructure in a portfolio

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 33页。

The benefits of adding investments in infrastructure assets to a portfolio most likely include:

将基础设施资产加入投资组合的好处最可能包括什么？

A	inflation protection only.	仅提供通胀保护。
B	low correlation with existing portfolio assets only.	仅与现有投资组合资产具有低相关性。
C	both inflation protection and low correlation with existing portfolio assets.	同时提供通胀保护，并与现有投资组合资产具有低相关性。

答案 / Supplied answer: C

中文解析

基础设施的收费或收入可能与通胀挂钩，因而可提供一定通胀保护；其现金流驱动因素也可能不同于现有资产，带来分散化作用。题目问可能的两类好处，因此A、B都遗漏一项。

Explanation

Infrastructure charges or revenues may be linked to inflation, offering some inflation protection. Different cash-flow drivers can also create diversification benefits. The question asks about both potential benefits, so A and B each omit one.

易错点 / Trap: 这是潜在组合好处，不表示所有项目都能完全抗通胀或始终保持低相关性。

These are potential benefits, not a guarantee of full inflation protection or permanently low correlation.

Q54 照付不议与需求风险

Take-or-pay and demand risk

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 54页。

With respect to infrastructure investments, a take-or-pay arrangement is most likely used to mitigate:

就基础设施投资而言，照付不议（take-or-pay）安排最可能用于缓解什么风险？

A	demand risk.	需求风险。
B	operational risk.	运营风险。
C	construction risk.	建设风险。

答案 / Supplied answer: A

中文解析

照付不议要求购买方按合同取用约定数量，或即使未取用也支付约定款项，从而减少项目收入对实际使用量的依赖。它主要缓解需求风险，并不直接解决运营故障、建设延误或成本超支。

Explanation

A take-or-pay arrangement requires the buyer to take the contracted quantity or pay an agreed amount even if it does not. This reduces revenue dependence on actual usage and chiefly mitigates demand risk, rather than operating failures, construction delays or cost overruns.

易错点 / Trap: 记住“用不用都要按约付款”，但仍须考虑买方履约能力。

Remember “take the contracted amount or pay”; buyer creditworthiness still matters.

单元 6 自然资源与商品

Natural Resources and Commodities

区分商品与生产商品的土地，再掌握持有成本与期货曲线。

Distinguish commodities from productive land, then learn cost of carry and the futures curve.

对应原题 / Original questions: Q14, Q17, Q51, Q48, Q53 | 5题 / questions

1. 商品与资源资产并不相同

1. Commodities and resource assets differ

概念 Concept	本题中的定位 Treatment in this set	原题 Questions
硬商品 Hard commodity	通常为开采所得的能源或矿产；原油属于硬商品。 Typically extracted energy or minerals; crude oil is a hard commodity.	Q14
商品本身 The commodity	牲畜是商品；林地、农业用地是生产资源的土地资产。 Livestock is a commodity; timberland and agricultural land are productive land assets.	Q17
林地收益 Timberland return	可以来自经营收入，也可以来自资产价值上升；不是二选一。 Can include an income stream and capital appreciation, not only one of the two.	Q51

2. 持有成本：利率 + 仓储 - 便利收益

2. Carry: financing + storage - convenience yield

$$F0(T) = S0 \times e^{[(r + u - y)T]}$$

S0 是现货价，F0(T) 是期限 T 的期货价；r 为无风险利率，u 为仓储成本率，y 为便利收益率。这里采用连续复利、可储存商品的标准持有成本表达，三项均为可比年化率。

S0 is spot, F0(T) is the futures price for maturity T, r is the risk-free rate, u is the storage cost rate and y is the convenience yield. This is the standard continuous-compounding carry expression for a storable commodity, using comparable annualized rates.

原答案使用单期简化式：期货价 ≈ 现货价×(1+r)+仓储成本-便利收益。它与上式的复利方式和成本表示不同，数值不能混用；但在Q48中，两种表达在成本与便利收益抵消后，均只看利率正负，结论同为C。

The source uses a simplified one-period relation: futures≈spot×(1+r)+storage costs-convenience yield. Its compounding and cost conventions differ from the formula above, so numerical inputs should not be mixed. In Q48 both reduce to the sign of the interest rate after storage and convenience yield offset, giving C.

Convenience yield（便利收益）是持有现货库存的非现金好处，例如保证生产不中断；它降低相对期货的持有成本。仓储费增加成本，便利收益抵消成本。不能把便利收益当成必须实际收到的利息。

Convenience yield is the non-cash benefit of holding physical inventory, such as avoiding a production interruption. Storage adds cost; convenience yield offsets it. It is not necessarily an interest payment received in cash.

Q48给出 u = y，两者抵消，剩下 F0(T) = S0 × e^(rT)。对正期限 T：r > 0 时期货价高于现货价；r = 0 时相等；r < 0 时低于现货。因此原题选 C。

In Q48, u = y, so these terms cancel and F0(T) = S0 × e^(rT). For positive T: futures exceed spot when r > 0, equal spot when r = 0, and fall below spot when r < 0. Therefore the answer is C.

3. 曲线形状与价格预测要分开

3. Separate curve shape from a price forecast

形状 Shape	含义 Meaning	原题 Question
正价差 Contango	本题口径：远期曲线向上倾斜，较远期限价格较高。 In this question, the forward curve slopes upward, with higher prices at longer maturities.	Q53
逆价差 Backwardation	作对照理解：远期曲线向下倾斜，较远期限价格较低。 For contrast, the forward curve slopes downward, with lower prices at longer maturities.	Q53对比 Q53 contrast

记忆抓手 / Recall cue

Contango 是同一时点不同到期合约的价格结构，不代表未来现货一定上涨，也不自动等于投资者获利。这里只讲原题需要的定价方向，不额外加入自编计算题。

Contango describes prices across maturities at one date. It does not guarantee a rise in future spot prices or an investor profit. The discussion covers the pricing direction needed by the supplied questions.

原题示例与逐题解析

Original questions and worked explanations

Q14 硬商品与软商品

Hard and soft commodities

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 14页。

Crude oil is categorized as:

原油被归类为：

A	a soft commodity.	软商品。
B	a hard commodity	硬商品。
C	neither a soft commodity nor a hard commodity.	既不属于软商品，也不属于硬商品。

答案 / Supplied answer: B

中文解析

硬商品主要包括通过开采或提取得到的能源、金属等资源；原油属于能源商品，因此选B。软商品通常指农业或畜牧类商品，原油不属于这一类。C错误地把原油排除在商品分类之外。

Explanation

Hard commodities include extracted resources such as energy products and metals. Crude oil is an energy commodity, so B is correct. Soft commodities generally cover agricultural or livestock products, which excludes crude oil. C incorrectly removes oil from these commodity categories.

易错点 / Trap: 原油看“开采”；农作物与畜牧产品看“种养”。

For oil, think extraction; for agricultural and livestock products, think growing or raising.

Q17 商品与生产性土地

Commodities versus productive land

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 17页。

Which of the following is best classified as a commodity?

以下哪一项最适合被归类为商品？

A	Livestock	牲畜。
B	Timberland	林地。
C	Agricultural land	农业用地。

答案 / Supplied answer: A

中文解析

牲畜属于可生产、交易及消费的商品，因此选A。林地和农业用地是可以持续产出木材或农产品的生产性土地资产；土地本身不等同于其产出的商品。B及C混淆了生产资料与产出物。

Explanation

Livestock is a produced, traded and consumable commodity, so A is correct. Timberland and agricultural land are productive land assets that can generate timber or agricultural output. The land itself is distinct from the commodities it produces; B and C confuse productive assets with their output.

易错点 / Trap: 林地≠木材；农地≠农产品。

Timberland is not timber; agricultural land is not agricultural output.

Q51 林地收益来源

Sources of timberland returns

来源 / Source: 题库 PDF 112页（章内 9页）；答案 PDF 51页。

Timberland investments offer:

林地投资提供什么？

A	an income stream only.	仅提供收入流。
B	the potential for capital gain only.	仅提供资本增值潜力。
C	both an income stream and the potential for capital gain.	既提供收入流，也提供资本增值潜力。

答案 / Supplied answer: C

中文解析

林地可通过木材采伐和销售产生收入，也可能因树木生长、木材价值变化及土地升值产生资本增值。因此两种收益来源都存在，A和B都遗漏一部分。

Explanation

Timberland can generate income through harvesting and timber sales. Biological growth, changes in timber values and land appreciation can also generate capital gains. Both sources exist, so A and B are incomplete.

易错点 / Trap: 区分林地这种生产性资产与已采伐的木材商品。

Distinguish productive timberland from harvested timber as a commodity.

Q48 持有成本模型

The cost-of-carry model

来源 / Source: 题库 PDF 112页（章内 9页）；答案 PDF 48页。

If a commodity's storage cost is equal to its convenience yield, its futures prices will be greater than its spot price if the risk-free rate is:

若某商品的储存成本等于其便利收益，在无风险利率为何值时，其期货价格会高于现货价格？

A	negative.	负值。
B	zero.	零。
C	positive.	正值。

答案 / Supplied answer: C

中文解析

按原答案的简化持有成本关系，期货价格 ≈ 现货价格×(1+r)+储存成本-便利收益，其中r为相同持有期的无风险利率。当储存成本与便利收益相抵，期货价格 ≈ 现货价格×(1+r)。因此r>0时，期货价格高于现货价格；r=0时相等；r<0时较低。

Explanation

Using the source’s simplified cost-of-carry relation, futures price≈spot price×(1+r)+storage costs-convenience yield, where r is the risk-free rate over the same holding period. When storage costs offset convenience yield, futures≈spot×(1+r). Thus r>0 gives futures above spot; r=0 gives equality; and r<0 gives futures below spot.

易错点 / Trap: 成本与便利收益相抵后，只剩融资成本决定本题的价差方向。

After storage cost offsets convenience yield, financing cost determines the price difference in this question.

Q53 期货升水结构

Contango

来源 / Source: 题库 PDF 113页（章内 10页）；答案 PDF 53页。

All else being equal, when a commodity futures market is in contango, the forward curve is most likely:

其他条件相同，当某商品期货市场处于期货升水结构（contango）时，远期曲线最可能呈现什么形态？

A	downward sloping.	向下倾斜。
B	flat.	平坦。
C	upward sloping.	向上倾斜。

答案 / Supplied answer: C

中文解析

本题中contango指较远到期合约价格高于较近到期合约价格，形成向上倾斜的期限曲线。向下倾斜对应backwardation；平坦曲线不符合本题所述升水结构。

Explanation

In this question, contango means more distant contracts trade above nearer contracts, producing an upward-sloping term structure. A downward-sloping curve is associated with backwardation, while a flat curve does not fit the stated contango condition.

易错点 / Trap: 这是不同到期日合约的价格结构，不是未来现货价格必然上涨的承诺。

This describes prices across contract maturities, not a guarantee that future spot prices will rise.

单元 7 对冲基金策略、流动性与估值

Hedge Fund Strategies, Liquidity and Valuation

先用策略关键词分类，再区分赎回限制，最后处理估值与尽职调查。

Classify strategies using their cues, distinguish redemption restrictions, and then address valuation and due diligence.

对应原题 / Original questions: Q26, Q20, Q21, Q39, Q42, Q44, Q7, Q30, Q13, Q47, Q12, Q59 | 12题 / questions

1. Hedge fund 不是单一的低风险策略

1. A hedge fund is not one low-risk strategy

对冲基金可以采用杠杆、衍生品以及多头和空头持仓。Q26把这种杠杆化私人投资工具识别为 hedge fund。名称中有"对冲"不代表收益不波动，也不代表市场 beta 必定为零。

Hedge funds may use leverage, derivatives, and long and short positions. Q26 identifies such a leveraged private investment vehicle as a hedge fund. The word hedge does not imply stable returns or zero market beta.

策略 Strategy	识别词与逻辑 Cue and logic	原题 Questions
股东积极主义 Activist	买入股权，试图影响公司政策或经营决策。 Buy equity to influence corporate policies or decisions.	Q20
事件驱动 Event-driven	围绕并购、重组或特殊事件；本题组以常见偏多头倾向描述。 Focus on mergers, restructuring or special events; the set describes a typical long bias.	Q21
宏观 Macro	从经济趋势与整体市场方向出发，采用top-down思路。 Take views on broad market direction based on economic trends; top-down analysis.	Q39
相对价值 Relative value	利用相关证券之间定价关系；可转债套利是代表。 Exploit pricing relationships among related securities; convertible bond arbitrage is an example.	Q42
市场中性 Market-neutral	通过组合安排尽量减少市场方向暴露；目标beta接近零。 Structure positions to minimize market exposure; target beta near zero.	Q44

Short-biased（偏空头）通常带负的市场方向暴露，fundamental long/short growth（基本面多空成长）仍可能带正的市场暴露。Market-neutral 的 beta 接近零只说明对指定市场因子的敏感度低，不代表总风险为零。

Short-biased strategies usually have negative market exposure. Fundamental long/short growth may retain positive exposure. A near-zero beta for a market-neutral strategy means low sensitivity to a specified market factor, not zero total risk.

2. 四种流动性安排

2. Four liquidity mechanisms

机制 Mechanism	何时起作用 When it applies	原题 Questions
锁定期 Lockup period	限制新投资者入场后一段时间内赎回；硬锁定禁止赎回，软锁定可能允许付费提前赎回。 Restricts early redemption after investment: a hard lockup prohibits it; a soft lockup may allow it for a fee.	Q7
通知期 Notice period	赎回之前须提前提出通知；强调通知须提前多久发出。 Requires advance notice before redemption; focuses on the lead time.	Q7 / Q30
赎回闸门 Gate	需要时临时限制可赎回规模，缓解集中赎回压力。 Temporarily limits redemptions when needed.	Q30
赎回费 Redemption fee	由退出者承担部分成本；按本题约定可增强剩余投资者回报。 Allocates costs to exiting investors and, under the set's convention, benefits remaining investors.	Q47

Q13问对冲基金最可能限制什么：与其常用的杠杆、衍生品相比，更典型的是限制赎回。限制赎回可以降低被迫出售低流动性资产的压力，但不改变资产本身的风险。

Q13 asks what hedge funds are most likely to restrict: redemptions, rather than leverage or derivatives that they may use extensively. Redemption restrictions can reduce forced-selling pressure but do not eliminate underlying asset risk.

3. 估值：越不可观察，越要审查

3. Valuation: less observable needs more scrutiny

Level 1 通常有活跃市场相同资产的报价；Level 2 主要使用其他可观察输入；Level 3 依赖重大不可观察输入、模型与管理判断。因此 Q12 选择 Level 3，需要最高程度尽职调查。

Level 1 generally uses quoted prices for identical assets in active markets. Level 2 uses other observable inputs. Level 3 relies on significant unobservable inputs, models and judgment. Q12 therefore selects Level 3 for the greatest due diligence.

Conservative close-out: Long → Bid; Short → Ask

多头要卖出，按较低买价 bid；空头要买回，按较高卖价 ask。相较使用反向报价或中间价，这样能更保守地估计净资产。对应 Q59。
A long position is sold at the lower bid; a short position is bought back at the higher ask. This is conservative close-out valuation. See Q59.

记忆抓手 / Recall cue

最常见混淆：lockup看入场后的赎回限制期；notice看走之前多久通知；gate看临时允许赎回的资金规模或比例。
Common confusion: lockup is the post-entry no-exit period; notice is the lead time before exit; a gate limits how much can exit temporarily.

原题示例与逐题解析

Original questions and worked explanations

Q26 对冲基金的基本识别

Identifying a hedge fund

来源 / Source: 题库 PDF 108页（章内 5页）；答案 PDF 26页。

A leveraged private investment vehicle that employs both long and short positions is most likely a:

一种使用杠杆、同时采用多头及空头持仓的私人投资工具，最可能是：

A	hedge fund.	对冲基金。
B	private equity fund.	私募股权基金。
C	venture capital fund.	风险投资基金。

答案 / Supplied answer: A

中文解析

杠杆与多空持仓结合，是对冲基金常见的投资安排，因此A最合适。私募股权基金主要投资非上市企业股权；风险投资是其中侧重早期企业的一类，两者通常不以持续构建可交易多空组合为主要特征。

Explanation

A leveraged vehicle combining long and short positions is most characteristic of a hedge fund, so A is correct. Private equity funds primarily invest in privately held companies; venture capital is the early-stage segment. Neither is typically defined by continually managing a tradable long/short portfolio.

易错点 / Trap: 私人投资工具不一定都是PE；看到杠杆、多头、空头组合，应先想到hedge fund。

A private vehicle is not automatically private equity; leverage plus long and short positions points toward a hedge fund.

Q20 积极股东策略

Activist strategy

来源 / Source: 题库 PDF 107页（章内 4页）；答案 PDF 20页。

A hedge fund strategy that seeks to influence a company's policies through the purchase of equity is best described as a(n):

一种对冲基金策略通过购买股权来影响公司的政策。这种策略最适合被称为：

A	activist strategy.	积极股东策略。
B	market-neutral strategy.	市场中性策略。
C	merger arbitrage strategy.	并购套利策略。

答案 / Supplied answer: A

中文解析

通过取得股权并向管理层或董事会施加影响，推动经营、资本配置或治理改变，是积极股东策略的核心，因此选A。市场中性策略重点在降低整体市场方向暴露；并购套利重点在交易宣布到完成期间的并购价差。

Explanation

Buying shares to influence management, capital allocation or governance is the defining feature of an activist strategy, so A is correct. Market-neutral strategies aim to reduce broad market directional exposure. Merger arbitrage focuses on price spreads associated with announced acquisitions and their completion.

易错点 / Trap: “买股份并影响公司决策”直接对应activist。

Buying equity to influence company decisions is the activist signal.

Q21 事件驱动策略的特征

Characteristics of event-driven strategies

来源 / Source: 题库 PDF 107页（章内 4页）；答案 PDF 21页。

Event-driven hedge fund strategies are most likely:

事件驱动型对冲基金策略最可能：

A	long biased	偏向做多。
B	based on 'top-down' analysis.	基于“自上而下”的分析。
C	exploiting short-term pricing discrepancies between two related securities.	利用两个相关证券之间的短期定价差异。

答案 / Supplied answer: A

中文解析

原答案选A：事件驱动策略常围绕并购、重组、破产或其他公司事件建立持仓，整体常呈做多倾向。B更贴近从经济趋势出发的宏观策略；C更贴近相对价值策略的表述。事件驱动策略也可能采用空头或对冲，不能把“偏多”理解为“只做多”。

Explanation

The supplied answer is A. Event-driven strategies position around mergers, restructurings, bankruptcies or other corporate events and often have a long bias. B more closely describes macro analysis. C is the standard description of a relative-value strategy. Event-driven funds may also short or hedge; a long bias does not mean long-only investing.

易错点 / Trap: Long biased表示整体倾向，不表示没有空头。事件驱动与相对价值可能有交集，但本题考典型归类。

Long biased describes a tendency, not the absence of short positions. Strategy categories can overlap, but this question tests their typical descriptions.

Q39 宏观对冲基金

Macro hedge funds

来源 / Source: 题库 PDF 110页（章内 7页）；答案 PDF 39页。

A hedge fund that seeks to profit from a view on overall market direction as influenced by economic trends best describes a:

某对冲基金根据经济趋势对整体市场方向形成判断，并试图从中获利。这最符合哪类基金？

A	macro hedge fund.	宏观对冲基金。
B	multi-strategy hedge fund.	多策略对冲基金。
C	market-neutral hedge fund.	市场中性对冲基金。

答案 / Supplied answer: A

中文解析

宏观基金以经济、政策等因素形成对利率、汇率、股票或商品市场方向的观点。多策略基金强调同时采用多种策略；市场中性基金则力求降低整体市场方向对收益的影响。

Explanation

Macro funds use economic and policy views to take positions on market direction across rates, currencies, equities or commodities. Multi-strategy funds combine several strategies, while market-neutral funds seek to reduce exposure to broad market direction.

易错点 / Trap: 看到overall market direction与economic trends，优先识别宏观策略。

Overall market direction and economic trends are key cues for a macro strategy.

Q42 相对价值策略

Relative value strategies

来源 / Source: 题库 PDF 111页（章内 8页）；答案 PDF 42页。

Which of the following is best described as a relative value hedge fund strategy?

以下哪项最符合相对价值型对冲基金策略？

A	Short biased	偏空策略。
B	Special situations	特殊情形策略。
C	Convertible bond arbitrage	可转换债券套利。

答案 / Supplied answer: C

中文解析

可转换债券套利利用可转债与相关股票或其他组成部分之间的相对定价偏差。偏空策略主要表达方向性判断；特殊情形策略通常围绕公司事件，属于事件驱动类别。

Explanation

Convertible bond arbitrage exploits relative mispricing between a convertible bond and its related equity or components. Short-biased strategies primarily express directional exposure. Special situations generally center on corporate events and fall within event-driven strategies.

易错点 / Trap: 相对价值的关键词是related securities与pricing discrepancy。

Relative value focuses on related securities and pricing discrepancies.

Q44 市场中性与贝塔

Market neutrality and beta

来源 / Source: 题库 PDF 111页（章内 8页）；答案 PDF 44页。

Which of the following hedge funds most likely have a beta close to zero?

以下哪类对冲基金最可能具有接近零的贝塔？

A	Short-biased funds	偏空基金。
B	Market-neutral funds	市场中性基金。
C	Fundamental long/short growth funds	基本面多空成长型基金。

答案 / Supplied answer: B

中文解析

市场中性基金通过配对或调整多空头寸，尽量抵消对整体市场的系统性暴露，使贝塔接近零。偏空基金通常具有负向市场暴露；多空成长型基金不一定对冲至市场中性。

Explanation

Market-neutral funds pair or adjust long and short positions to offset broad market exposure, targeting beta near zero. Short-biased funds generally have negative market exposure. Fundamental long/short growth funds do not necessarily neutralize market beta.

易错点 / Trap: 贝塔接近零只代表市场方向暴露较低，不代表没有风险或不会亏损。

Near-zero beta means limited market-direction exposure, not zero risk or guaranteed profits.

Q7 锁定期、通知期与赎回门槛

Lockups, notice periods and gates

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 7页。

A measure that restricts new investors in a hedge fund from redeeming their capital for a set amount of time in order to implement the fund's investment strategy is known as a:

为了实施基金的投资策略，一项措施限制对冲基金的新投资者在一段规定时间内赎回其资本。这项措施称为：

A	gate.	赎回限制机制（gate）。
B	notice period.	赎回通知期。
C	lockup period.	锁定期。

答案 / Supplied answer: C

中文解析

锁定期从投资进入基金后开始，在规定期限内限制新投资者赎回，因此选C。通知期是实际赎回前必须提前通知基金的时间；gate则是在需要时限制某一期可赎回的金额或比例。题干的关键词是“新投资者”及“规定时间内不得赎回”。

Explanation

A lockup restricts a new investor's redemption rights for a specified period after entry, so C is correct. A notice period requires advance notice before redemption. A gate limits the amount or proportion redeemable during a period when invoked. The decisive phrases are "new investors" and "for a set amount of time."

易错点 / Trap: Lockup看进入后的锁定时间；notice看提前通知；gate看当期赎回限制。

Lockup concerns time after entry; notice concerns advance warning; gate concerns limits on current redemptions.

Q30 Gate赎回限制

Redemption gates

来源 / Source: 题库 PDF 109页（章内 6页）；答案 PDF 30页。

Which of the following hedge fund mechanisms is most likely used to impose a temporary restriction on redemptions if needed?

以下哪一种对冲基金机制最可能用于在有需要时暂时限制赎回？

A	Gate	赎回限制机制（gate）。
B	Notice period	赎回通知期。
C	Lockup period	锁定期。

答案 / Supplied answer: A

中文解析

Gate允许基金在需要时限制某一期可赎回的资本金额或比例，因此选A。通知期要求投资者提前提出赎回通知；锁定期则在初始投资后的一段约定期间内限制赎回。题干强调“在有需要时暂时施加限制”，与gate对应。

Explanation

A gate permits the fund to restrict the amount or proportion of capital redeemable during a period when necessary, so A is correct. A notice period requires advance redemption notice; a lockup restricts withdrawals for a contractual period after initial investment. The decisive wording is a temporary restriction imposed when needed.

易错点 / Trap: Q7考进入后锁定；Q30考必要时临时限制，两题须一起区分。

Q7 tests a post-entry lockup; Q30 tests a temporary limit invoked when needed. Learn the distinction together.

Q13 对冲基金的赎回限制

Hedge fund redemption restrictions

来源 / Source: 题库 PDF 106页（章内 3页）；答案 PDF 13页。

Hedge funds are most likely to place restrictions on:

对冲基金最可能对以下哪一项施加限制？

A	redemption s.	赎回。
B	the use of leverage.	杠杆的使用。
C	the use of derivatives.	衍生品的使用。

答案 / Supplied answer: A

中文解析

对冲基金常通过锁定期、通知期或gate管理赎回，以避免被迫出售缺乏流动性的持仓，因此选A。使用杠杆及衍生品通常是其策略灵活性的一部分。B及C并不意味着完全没有限制，但在本题的一般性比较中，赎回限制最具代表性。

Explanation

Hedge funds commonly manage redemptions through lockups, notice periods or gates to reduce forced sales of illiquid positions, so A is correct. Leverage and derivatives are often part of their strategic flexibility. B and C are not necessarily unrestricted, but redemption restrictions are the characteristic feature emphasized by this question.

易错点 / Trap: 题目问“最可能的典型限制”，并非断言杠杆和衍生品永远没有约束。

The question asks for a characteristic restriction, not a claim that leverage and derivatives are never constrained.

Q47 赎回费与留存投资者

Redemption fees and remaining investors

来源 / Source: 题库 PDF 112页（章内 9页）；答案 PDF 47页。

What is the most likely effect of a redemption fee on the returns of the remaining investors in a hedge fund? A redemption fee:

赎回费最可能如何影响对冲基金中剩余投资者的回报？赎回费会：

A	reduces investor returns.	降低投资者回报。
B	has no effect on investor returns.	不影响投资者回报。
C	enhances investor returns.	提高投资者回报。

答案 / Supplied answer: C

中文解析

题库所指赎回费通常支付给基金本身，增加留存于基金的资产，保护剩余投资者免受赎回引发的交易及流动性成本影响。因此，相对于没有该费用的情形，留存投资者的回报受到保护并提高；不能把它理解为仅支付给管理人的额外报酬。

Explanation

The source’s redemption fee is normally paid to the fund itself, adding to retained fund assets and protecting remaining investors against transaction and liquidity costs caused by redemptions. It enhances their returns relative to the situation without the fee; it should not be treated simply as extra compensation paid to the manager.

易错点 / Trap: 分清支付赎回费的退出者与受到保护的留存投资者。

Distinguish the redeeming investor who pays the fee from the remaining investors who benefit.

Q12 估值层级与尽职调查

Valuation hierarchy and due diligence

来源 / Source: 题库 PDF 105页（章内 2页）；答案 PDF 12页。

Which of the following hedge fund investments require the highest level of scrutiny and due diligence?

以下哪一种对冲基金投资需要最高程度的审查和尽职调查？

A	Level 1 assets	第1层级资产。
B	Level 2 assets	第2层级资产。
C	Level 3 assets	第3层级资产。

答案 / Supplied answer: C

中文解析

第3层级估值依赖重大不可观察输入，对模型、假设及管理者判断的依赖较大，因而需要最严格的审查，选C。第1层级有相同资产在活跃市场的未调整报价；第2层级主要采用其他可观察输入。这里比较的是估值可验证性，不能简单理解为层级数字越高，资产必然亏损越大。

Explanation

Level 3 valuations rely on significant unobservable inputs and therefore on models, assumptions and judgment, requiring the most scrutiny. C is correct. Level 1 uses unadjusted quoted prices for identical assets in active markets; Level 2 relies on other observable inputs. The hierarchy concerns valuation observability, not a guarantee that higher-numbered assets will generate larger losses.

易错点 / Trap: Level 3 = 不可观察输入较重要 = 估值和尽调压力最大。

Level 3 = significant unobservable inputs = the greatest valuation and due-diligence challenge.

Q59 保守估值的买卖价

Bid-ask prices and conservative valuation

来源 / Source: 题库 PDF 114页（章内 11页）；答案 PDF 59页。

Which of the following is the most conservative approach to valuing a hedge fund's underlying positions?

以下哪一种是对对冲基金底层头寸进行估值时最保守的方法？

A	Using bid prices for long positions and ask prices for short positions	多头头寸用买入价（bid），空头头寸用卖出价（ask）。
B	Using bid prices for short positions and ask prices for long positions	空头头寸用买入价（bid），多头头寸用卖出价（ask）。
C	Using the average of the bid and ask prices for both long and short positions	多头和空头头寸都使用买入价与卖出价的平均值。

答案 / Supplied answer: A

中文解析

多头平仓需要卖出，只能按市场买价bid估算可变现金额；空头平仓需要买回，应按较高的市场卖价ask估算回补成本。这使资产估值偏低、空头负债估值偏高，更为保守。B相反；C的中间价未反映完整平仓价差。

Explanation

Closing a long position requires selling at the bid. Closing a short requires buying back at the ask. This lowers long-asset values and raises short-covering liabilities, making A more conservative. B reverses the logic, while C's midpoint valuation does not reflect the full liquidation spread.

易错点 / Trap: bid/ask是市场报价方的买卖方向；从持有人角度，多头卖给bid，空头买回ask。

Bid and ask describe the quoting market's side: sell a long at the bid and cover a short at the ask.

单元 8 数字资产：按原题语境学习

Digital Assets in the Supplied Question Context

掌握发行人、投资路径与价格驱动，同时识别原资料中过度概括的表述。

Learn issuers, investment routes and return drivers while recognizing overgeneralized statements in the source.

对应原题 / Original questions: Q5, Q2, Q4, Q3, Q6, Q1 | 6题 / questions

1. 发行与持有路径

1. Issuance and holding routes

加密货币可由个人、公司或组织发行。直接持有通常涉及加密钱包；通过 ETF / 信托等产品取得敞口，是对该产品的投资，投资者通常不必自己保管底层加密资产私钥。Q5选 C，Q2选 A。

Cryptocurrencies can be issued by individuals, corporations or organizations. Direct holdings typically involve a crypto wallet. Exposure through an ETF or trust is an investment in that vehicle; investors generally do not need to manage the underlying crypto private keys. Q5 is C and Q2 is A.

> 理解提示 / Context note
>
> Q2原解析谈到ETF通常不直接持币，这不是今天可直接通用的判断。美国SEC已于2024年1月10日批准现货比特币ETP上市交易。仍应抓住本题的有效区别：投资者直接持币与持有产品份额的保管方式不同。[S2]
>
> Q2's supplied explanation says ETFs typically do not hold crypto directly; that is not a universal present-day statement. The US SEC approved spot bitcoin ETP listings on 10 January 2024. The useful distinction remains how investors hold direct coins versus product shares. [S2]

2. 收益驱动与相关性

2. Return drivers and correlation

Q4把市场采用程度提高 increased market adoption 识别为比特币回报的重要驱动；题目不把其视作有合同利息、股息或企业现金流支持的证券。Q3的原答案是相关性 increasing，但这是原资料的历史描述，不能背成所有时期一直上升的规律。

Q4 identifies increased market adoption as a major driver of bitcoin returns; it does not treat bitcoin as a security supported by contractual interest, dividends or corporate cash flow. Q3's supplied answer is increasing correlation, a historical description in the source rather than a rule that correlation rises in every period.

3. 数字资产范围大于无资产支持的加密币

3. Digital assets extend beyond unbacked crypto

> 理解提示 / Context note
>
> Q6原答案为 C，但题干把 digital assets 写得很宽，原解析实际使用 most digital assets。更稳妥的理解对象是本题关注的不由底层资产或合同现金流支持的加密资产；不能把"所有数字资产都没有底层资产 / 现金流价值"当成定义。代币化债券等资产并不适用该绝对表述。
>
> The supplied answer to Q6 is C. The stem broadly says digital assets, while the explanation says most digital assets. Read it in the context of unbacked crypto without underlying asset claims or contractual cash flows. Do not define all digital assets as lacking asset or cash-flow backing; tokenized bonds are a counterexample.

4. 去中心化不等于绝对安全

4. Decentralization is not absolute security

Q1按原答案选 A：去中心化交易所较难由单一中心控制和监管，并可减少某些单点故障。但原解析关于攻击几乎必然失败的说法过强。智能合约漏洞仍可能被利用，因此不能得出“DEX不会被攻击”或“一定比所有中心化交易所安全”的现实结论。[S3]

Q1’s supplied answer is A: decentralized exchanges are harder to control or regulate through one center and may reduce certain single points of failure. The source’s claim that attacks are almost certain to fail is too strong. Smart-contract vulnerabilities can be exploited, so do not infer that DEXs cannot be hacked or are always safer than all centralized exchanges. [S3]

记忆抓手 / Recall cue

本单元答题方式：先记录原答案，再注明成立语境。概念校核用于防止误学，不改变用户提供的题干、选项或答案记录。
Study method: record the supplied answer and its scope. Concept checks prevent overgeneralization without changing the supplied stem, choices or answer record.

原题示例与逐题解析

Original questions and worked explanations

Q5 加密货币的发行主体

Cryptocurrency issuers

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 5页。

Cryptocurrencies can be issued by:

加密货币可以由以下哪些主体发行？

A	individuals only.	仅个人。
B	corporations and organisations only.	仅公司及组织。
C	individuals, corporations and organisations.	个人、公司及组织。

答案 / Supplied answer: C

中文解析

在本题讨论的技术和组织形式范围内，个人、公司或其他组织均可创建或发行加密货币，因此C最完整。A及B都不当地把发行主体限制为其中一部分。发行能力本身不代表获得监管批准，也不代表具有价值或信用保障。

Explanation

Within the technical and organizational scope of the question, individuals, corporations and other organizations can create or issue cryptocurrencies. C is therefore the most complete answer; A and B exclude eligible categories. The ability to issue a token does not itself establish regulatory approval, economic value or credit protection.

易错点 / Trap: 可以发行，不等于发行经过批准或投资价值有保证。
Being able to issue a token does not imply approval or guaranteed investment value.

Q2 直接持有与间接持有数字资产

Direct and indirect digital asset ownership

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 2页。

Which of the following forms of digital asset investment most likely involves the use of a cryptocurrency wallet?

以下哪一种数字资产投资形式最可能涉及使用加密货币钱包？

A	Direct investment	直接投资
B	Indirect investment via ETF s	通过ETF间接投资
C	Indirect investment via coin trusts	通过加密货币信托间接投资

答案 / Supplied answer: A

中文解析

直接持有加密货币需要通过钱包及密钥安排访问或转移资产，因此选A。通过ETF或信托投资时，投资者持有的是基金或信托份额，通常通过证券账户交易，不需要自己管理底层加密资产的钱包。基金自身是否直接持币，与份额持有人是否需要钱包，是两个不同问题。

Explanation

Direct cryptocurrency ownership involves wallet and key arrangements for access and transfers, so A is correct. An ETF or trust investor owns shares, normally traded through a securities account, and need not personally manage the underlying cryptocurrency wallet. Whether the vehicle itself holds coins directly is separate from whether its shareholder needs a wallet.

易错点 / Trap: 识别"投资者持有什么"：币对应钱包；基金或信托份额对应证券账户。

Identify what the investor owns: coins involve wallets; fund or trust shares are held through securities accounts.

Q4 比特币收益驱动因素

Drivers of bitcoin returns

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 4页。

Which of the following is most likely a major driver of bitcoin returns?

以下哪一项最可能是比特币收益的主要驱动因素？

A	Increased market adoption	市场接受度提高
B	The prospect of underlying cash flow generation	底层现金流创造能力的前景
C	Consistently high correlation with traditional asset classes	与传统资产类别持续保持较高相关性

答案 / Supplied answer: A

中文解析

市场采用和接受度提高可增加对比特币的需求，因此A最合适。比特币本身不赋予持有人对发行企业经营现金流的索取权，B不适用。相关性描述共同变动关系，并非稳定不变的独立价值来源，因此C不能作为本题主要驱动因素。

Explanation

Greater adoption can increase demand for bitcoin, making A the best answer. Bitcoin itself does not grant a claim on an issuing company's operating cash flows, so B is inappropriate. Correlation describes co-movement; it is neither necessarily stable nor an independent source of value, so C is not the primary driver identified here.

易错点 / Trap: 比特币的需求与采用率，不等于股票的股利或经营现金流。

Bitcoin demand and adoption are different from equity dividends or operating cash flows.

Q3 加密资产相关性的时间变化

Time variation in cryptocurrency correlations

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 3页。

The correlation of cryptocurrencies with traditional assets has been:

加密货币与传统资产之间的相关性呈现以下哪种变化？

A	decreasing.	下降。
B	steady.	稳定。
C	increasing.	上升。

答案 / Supplied answer: C

中文解析

原答案选C，反映资料所讨论时期中加密货币与传统市场联系增强的现象。A与该历史观察相反；B错误地认为相关性保持不变。原题没有给出观察期间、资产范围或估计方法，不能据此推导相关性在所有年份、所有市场状态下都持续上升。

Explanation

The supplied answer is C, reflecting stronger links with traditional markets in the period discussed by the source. A contradicts that observation; B assumes an unchanged relationship. The question gives no observation window, asset universe or estimation method, so it cannot establish a continuously rising correlation across all periods and market conditions.

易错点 / Trap: 这是资料中的历史判断，不能当作永恒规律，也不能直接等同于今天的相关系数。

Treat this as a historical statement in the source, not a permanent rule or an estimate of today's correlation.

Q6 数字资产的价值基础：题库表述边界

Digital asset value: limits of the source wording

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 6页。

Compared to traditional financial assets, digital assets:

与传统金融资产相比，数字资产：

A	can be invested in through indirect investment vehicles such as ETF s.	可以通过ETF等间接投资工具进行投资。
B	are generally recorded in private ledgers maintained by central intermediaries.	一般记录在由中央中介机构维护的私人账本中。
C	do not have an inherent value based on underlying assets or on potential cash flows.	不具有基于底层资产或潜在现金流的内在价值。

答案 / Supplied answer: C

中文解析

原答案选C，所采用的是比特币等不附带底层资产或现金流索取权的加密资产语境。A也适用于传统资产，因此不是这里要找的区别；B描述中央中介维护账本，而非题库强调的分布式记录。原题把“数字资产”概括得过宽：代币化债券、资产支持代币等可能有底层资产或现金流，不能把C推广到全部数字资产。

Explanation

The supplied key selects C in the context of cryptocurrencies such as bitcoin that do not carry claims on underlying assets or cash flows. A also applies to traditional assets, so it does not establish the intended distinction. B describes centralized recordkeeping rather than the distributed records emphasized by the source. The wording is too broad for all digital assets: tokenized bonds and asset-backed tokens may have underlying assets or cash flows.

易错点 / Trap: 考试按原题语境识别C；理解时限定资产类型，勿把“数字化形式”等同于“没有现金流”。

Identify C under the source's intended context, but do not equate a digital format with an absence of cash flows.

Q1 去中心化交易所与中心化交易所

Decentralized versus centralized exchanges

来源 / Source: 题库 PDF 104页（章内 1页）；答案 PDF 1页。

Compared to centralized cryptocurrency exchanges, decentralized exchanges are:

与中心化加密货币交易所相比，去中心化交易所：

A	less likely to be regulated and less susceptible to attacks from hackers.	较不可能受到监管，也较不容易遭受黑客攻击。
B	less likely to be regulated and more susceptible to attacks from hackers.	较不可能受到监管，但较容易遭受黑客攻击。
C	more likely to be regulated and less susceptible to attacks from hackers.	较可能受到监管，但较不容易遭受黑客攻击。

答案 / Supplied answer: A

中文解析

原答案选A：题目强调去中心化架构没有单一中央控制点，因此在本题比较框架下，受到集中监管及单点攻击的可能性较低。B的攻击方向相反；C的监管方向相反。理解边界：这是资料采用的简化比较，不代表去中心化交易所不受监管或不会被攻击；智能合约、跨链桥及其他环节仍可能存在漏洞。

Explanation

The supplied key selects A. Under the question's simplified framework, decentralized architecture lacks a single central control point, reducing centralized oversight and single-point vulnerability. B reverses the security comparison; C reverses the regulatory comparison. This does not mean decentralized exchanges are unregulated or immune to attack: smart contracts, bridges and other components can still contain vulnerabilities.

易错点 / Trap: 保留题库答案，同时把“没有单点故障”与“没有安全风险”分开。

Retain the supplied answer, but distinguish the absence of a single point of failure from the absence of security risk.

附录A 费用公式与六题核算

Appendix A: fee formulas and six checked cases

计算口径与单元2相同：管理费按年末费前AUM，业绩费先扣管理费，再按题目扣除硬门槛。所有金额单位为million（百万）；Q49为EUR，其余为USD。没有门槛的题，H=0。

Use the Unit 2 convention: management fee on ending pre-fee AUM, incentive fee net of the management fee, and a hard hurdle where stated. All amounts are in millions: EUR for Q49, USD otherwise. H=0 when no hurdle is given.

题号 Q	期初 A0	费前期末 A1	管理费 M	门槛额 H	业绩费 I	总费用 M+I	题目所求 Requested result
Q19	500	610	12.20	50	7.17	19.37	业绩费7.17≈7.2；A Incentive fee 7.17≈7.2; A
Q24	200	250	5.00	0	9.00	14.00	总费用14；B Total fees 14; B
Q45	100	120*	1.20	3	2.37	3.57	净回报16.43%≈16.4%；B Net return 16.43%≈16.4%; B
Q46	1,500	1,800	36.00	90	34.80	70.80	总费用70.8；B Total fees 70.8; B
Q49	200	230	3.45	10	3.31	6.76	总费用6.76≈6.8；A Total fees 6.76≈6.8; A
Q61	50	60	1.20	0	1.76	2.96	净回报14.08%；B Net return 14.08%; B

* Q45原题印作S120 million；此表按答案文件明确的$120 million核算。表中的其他衍生数字为对原题数据的计算结果，没有更改原题。

* Q45 prints S120 million; the table follows the answer file's explicit $120 million. Other derived figures are calculations from the original data, not changes to the questions.

$A1 = A0(1+g);\ M = mA1;\ H = hA0;\ I = p \times \max(0, A1-A0-M-H)$

$Total = M+I;\ Rnet = (A1-A0-Total)/A0$

看到的词 Wording	立即采取的动作 Action
End-of-year AUM before fees	先计算期末费前AUM，管理费乘这个金额。 Use ending gross AUM as the management-fee base.
Net of management fees	算业绩费之前先减管理费。 Deduct the management fee before calculating incentives.
Hard hurdle / above a hurdle	用期初资本把门槛收益率换成金额，再扣除。 Convert the hurdle to an amount using beginning capital, then deduct it.
Total fees / incentive fee / net return	分别对应M+I、I、净利润÷A0，不要答错所求量。 Return M+I, I, or net profit/A0 respectively.

附录B 考前知识速查

Appendix B: rapid concept review

遮住右侧说明，看到术语后先用自己的话复述，再返回对应原题检查。这里没有加入新题；所有提示都指向已整理的原题。

Cover the explanation, restate each term in your own words, then revisit the linked original question. These are recall prompts for the supplied material, not new questions.

关键词 Term	必须能说出的规则 Rule to recall	原题 Questions
传统投资 Traditional assets	股票、债券和现金的多头；本题现金属传统。 Long-only stocks, bonds and cash; cash is traditional here.	Q23 / Q55
控制与专业要求 Control and expertise	基金通常最低，直接投资通常最高。 Generally lowest for fund investing, highest for direct investing.	Q16 / Q32
共同投资 Co-investing	有权与基金一起直接投资，不是义务。 The right, not an obligation, to invest directly alongside the fund.	Q34
管理费基数 Management fee base	先看期初或期末、AUM或承诺资本。 Check beginning/end, AUM/committed capital.	Q25 / Q37
硬门槛计算 Hard hurdle calculation	先扣管理费，再扣门槛金额，只对超额利润收费。 Deduct management fee and hurdle amount before charging excess profit.	Q19 / Q49
高水位 High-water mark	防同一段业绩重复收费。 Prevents charging twice on the same performance.	Q27
整体分配与追回 Whole-fund / clawback	整体分配较有利LP；追回多收业绩费不等于保本。 Whole-fund generally favors LPs; fee clawback is not capital protection.	Q29 / Q58
风险投资与成长资本 VC and growth capital	VC属私募股权；成长资本常为少数股权。 VC is private equity; growth capital often uses minority equity.	Q8 / Q11
夹层债务 Mezzanine debt	优先债务之后、股权之前，可能带认股权证。 Below senior debt, above equity; may have warrants.	Q36
房地产 Real estate	异质、高交易成本；直接持有需管理时间。 Heterogeneous, costly to transact, time-consuming to manage.	Q35 / Q43 / Q56
Core与Value-add Core and value-add	优质成熟物业为Core；出租空置及重定位为Value-add。 Established quality property is core; vacancy leasing and repositioning are value-add.	Q40 / Q52
基础设施阶段 Infrastructure stage	Greenfield建设风险较高；Brownfield有运营记录。 Greenfield has development risk; brownfield has operating history.	Q18 / Q31
照付不议 Take-or-pay	主要缓解需求风险。 Primarily mitigates demand risk.	Q54
持有成本 Cost of carry	r + u - y；u=y时，只需判断r正负。 r + u - y; when u=y, inspect the sign of r.	Q48

关键词 Term	必须能说出的规则 Rule to recall	原题 Questions
Contango Contango	远期曲线向上，不等于未来现货必涨。 Upward curve; not a promise that future spot will rise.	Q53
市场中性 Market-neutral	beta接近零不代表总风险为零。 Beta near zero does not mean zero total risk.	Q44
流动性条款 Liquidity terms	Lockup锁定；notice提前通知；gate临时限制。 Lockup restricts early exit; notice sets lead time; gate temporarily restricts.	Q7 / Q30
保守估值 Conservative valuation	多头bid，空头ask；Level 3审查最多。 Long at bid, short at ask; Level 3 needs most scrutiny.	Q12 / Q59
数字资产 Digital assets	直接持币涉及钱包；Q1/Q2/Q3/Q6须按语境理解。 Direct coins involve wallets; qualify Q1/Q2/Q3/Q6 by context.	Q1-Q6

附录C 61题答案与定位索引

Appendix C: answer and location index

答案全部来自第二份文件。本讲义页码可用于打印查阅；点击题号可直接跳转。最后一列依次为第一份题库的总PDF页码与第二份答案PDF页码。

All keys come from the second supplied PDF. Guide-page numbers support printed lookup; question numbers are clickable. The last column lists the full question-bank PDF page followed by the answer-file PDF page.

* Q9的A是原答案，定义口径存在争议，必须连同该题的说明一起理解。

* Q9 retains the supplied A; its definition is disputed and must be read with the qualification.

原题 / 答案 Question / key	知识点定位 Study-unit location	本讲义页 Guide p.	题库 / 答案页 Source / key p.
Q1 / A	单元 8 / Unit 8 Decentralized versus centralized exchanges	53	104 / 1
Q2 / A	单元 8 / Unit 8 Direct and indirect digital asset ownership	51	104 / 2
Q3 / C	单元 8 / Unit 8 Time variation in cryptocurrency correlations	52	104 / 3
Q4 / A	单元 8 / Unit 8 Drivers of bitcoin returns	51	104 / 4
Q5 / C	单元 8 / Unit 8 Cryptocurrency issuers	50	104 / 5
Q6 / C	单元 8 / Unit 8 Digital asset value: limits of the source wording	52	104 / 6
Q7 / C	单元 7 / Unit 7 Lockups, notice periods and gates	46	105 / 7
Q8 / C	单元 3 / Unit 3 Private equity categories	25	105 / 8
Q9 / A*	单元 3 / Unit 3 Leveraged loans: source and market terminology differ	27	105 / 9
Q10 / B	单元 3 / Unit 3 Exit through a SPAC	26	105 / 10
Q11 / C	单元 3 / Unit 3 Growth capital	25	105 / 11
Q12 / C	单元 7 / Unit 7 Valuation hierarchy and due diligence	48	105 / 12
Q13 / A	单元 7 / Unit 7 Hedge fund redemption restrictions	47	106 / 13

原题 / 答案 Question / key	知识点定位 Study-unit location	本讲义页 Guide p.	题库 / 答案页 Source / key p.
Q14 / B	单元 6 / Unit 6 Hard and soft commodities	38	106 / 14
Q15 / A	单元 1 / Unit 1 Control in co-investing and direct investing	9	106 / 15
Q16 / C	单元 1 / Unit 1 Flexibility of investment methods	8	106 / 16
Q17 / A	单元 6 / Unit 6 Commodities versus productive land	39	106 / 17
Q18 / B	单元 5 / Unit 5 Greenfield infrastructure investing	35	106 / 18
Q19 / A	单元 2 / Unit 2 Hard hurdle and incentive fees net of management fees	17	107 / 19
Q20 / A	单元 7 / Unit 7 Activist strategy	43	107 / 20
Q21 / A	单元 7 / Unit 7 Characteristics of event-driven strategies	44	107 / 21
Q22 / B	单元 5 / Unit 5 Social and economic infrastructure	34	107 / 22
Q23 / B	单元 1 / Unit 1 Definition and typical features of alternatives	7	107 / 23
Q24 / B	单元 2 / Unit 2 Total fees without a hurdle	16	108 / 24
Q25 / A	单元 2 / Unit 2 Management-fee bases for hedge funds and private equity	14	108 / 25
Q26 / A	单元 7 / Unit 7 Identifying a hedge fund	43	108 / 26
Q27 / C	单元 2 / Unit 2 High-water mark	21	108 / 27
Q28 / C	单元 1 / Unit 1 Joint ventures and direct real estate	10	108 / 28
Q29 / C	单元 2 / Unit 2 Whole-of-fund waterfall	21	108 / 29
Q30 / A	单元 7 / Unit 7 Redemption gates	46	109 / 30

原题 / 答案 Question / key	知识点定位 Study-unit location	本讲义页 Guide p.	题库 / 答案页 Source / key p.
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Q32 / B	单元 1 / Unit 1 Investment methods and expertise	8	109 / 32
Q33 / C	单元 5 / Unit 5 Infrastructure in a portfolio	36	109 / 33
Q34 / A	单元 1 / Unit 1 Co-investment rights	9	109 / 34
Q35 / C	单元 4 / Unit 4 Burden of direct real estate ownership	31	109 / 35
Q36 / C	单元 3 / Unit 3 Mezzanine debt and the capital structure	27	110 / 36
Q37 / B	单元 2 / Unit 2 The private equity J-curve	22	110 / 37
Q38 / A	单元 3 / Unit 3 Private equity exits	26	110 / 38
Q39 / A	单元 7 / Unit 7 Macro hedge funds	44	110 / 39
Q40 / C	单元 4 / Unit 4 Core real estate	32	110 / 40
Q41 / A	单元 1 / Unit 1 Co-investing and fees	10	110 / 41
Q42 / C	单元 7 / Unit 7 Relative value strategies	45	111 / 42
Q43 / A	单元 4 / Unit 4 Heterogeneity of real estate	30	111 / 43
Q44 / B	单元 7 / Unit 7 Market neutrality and beta	45	111 / 44
Q45 / B	单元 2 / Unit 2 Net return with a hurdle	19	111 / 45
Q46 / B	单元 2 / Unit 2 Total fees versus incentive fees	20	111 / 46
Q47 / C	单元 7 / Unit 7 Redemption fees and remaining investors	47	112 / 47

原题 / 答案 Question / key	知识点定位 Study-unit location	本讲义页 Guide p.	题库 / 答案页 Source / key p.
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Q49 / A	单元 2 / Unit 2 Hard hurdles and net-of-management fees	18	112 / 49
Q50 / B	单元 3 / Unit 3 Venture debt	28	112 / 50
Q51 / C	单元 6 / Unit 6 Sources of timberland returns	39	112 / 51
Q52 / B	单元 4 / Unit 4 Value-add real estate	32	113 / 52
Q53 / C	单元 6 / Unit 6 Contango	40	113 / 53
Q54 / A	单元 5 / Unit 5 Take-or-pay and demand risk	36	113 / 54
Q55 / B	单元 1 / Unit 1 The boundary of traditional investments	6	113 / 55
Q56 / A	单元 4 / Unit 4 Private real estate market characteristics	31	113 / 56
Q57 / C	单元 1 / Unit 1 Tangible and intangible alternative assets	7	113 / 57
Q58 / C	单元 2 / Unit 2 Clawback provisions	22	114 / 58
Q59 / A	单元 7 / Unit 7 Bid-ask prices and conservative valuation	48	114 / 59
Q60 / C	单元 3 / Unit 3 Relative risk of private debt	28	114 / 60
Q61 / B	单元 2 / Unit 2 Net return without a hurdle	15	114 / 61

附录D 原资料勘误与适用语境

Appendix D: source corrections and context

以下记录区分三类情况：明确笔误、术语口径争议、时间或适用范围限制。原题和原答案不被偷偷改写；讲义中的计算和解释明确说明处理方式。

The notes distinguish explicit typos, terminology disputes, and time or scope limitations. Original questions and keys are not silently changed; calculations and explanations state how each issue is handled.

Q19 公式笔误 / Formula typo

原解析写500×22%=610，漏了1+；应为500×(1+22%)=610。后续正确业绩费7.17百万，原答案A不变。

The source prints 500×22%=610, omitting 1+. The correct expression is 500×(1+22%)=610. Incentive fee remains 7.17 million and the supplied A is retained.

Q24 费用名称笔误 / Fee-label typo

原解析把扣除的5百万美元称为incentive，实际是management fee。管理费5加业绩费9，总费用14百万，原答案B不变。

The source calls the deducted USD 5 million an incentive fee; it is the management fee. Management 5 plus incentive 9 equals total fees 14 million. B is retained.

Q45 金额符号异常 / Currency-symbol issue

原题确实印作S120 million。英文保留原样；中文和计算明确按答案文件使用的$120 million处理，净回报16.43%，原答案B。

The question actually prints S120 million. English is preserved; translation and calculations explicitly use the answer file’s $120 million. Net return is 16.43%; supplied answer B.

Q9 定义口径争议 / Definition dispute

原答案A采用贷款提供方加杠杆的口径；常见市场定义侧重借款人高负债和 / 或低评级。保留A并加警示，不能擅改B，也不能把A当作普遍定义。[S1]

The supplied A concerns leverage at the lender. Common market usage instead emphasizes high borrower debt and/or low ratings. Retain A with a qualification, without changing it to B or treating A as universal. [S1]

Q1 安全性过度概括 / Security overstatement

分布式架构可以减少某些单点故障，但无法保证攻击失败。原答案A按本题简化框架保留；现实仍有智能合约等安全风险。[S3]

Distributed architecture can reduce certain single points of failure but cannot guarantee failed attacks. A is retained under the source’s simplified framework; smart-contract and other security risks remain. [S3]

Q2 历史语境 / Historical context

原解析关于ETF通常不直接持币的说法不能通用于现行产品。SEC于2024-01-10批准现货比特币ETP。本题仍考投资者直接持币是否涉及钱包，原答案A。[S2]

The source’s statement that ETFs typically do not hold coins directly is not universal today. The SEC approved spot bitcoin ETPs on 2024-01-10. The wallet question still distinguishes direct investor holdings; A is retained. [S2]

Q3 样本期缺失 / Missing sample period

原答案C为资料描述的相关性上升；原题未给观察期或统计方法，不据此推定各时期一直上升。

The supplied C describes rising correlation in the source. No sample period or method is given, so it is not a claim of an increase in every period.

Q6 适用范围过宽 / Overbroad scope

选项C说digital assets，原解析却说most digital assets。按所指非资产支持型加密资产理解，原答案C；不推广到代币化债券等。

Choice C says digital assets while the explanation says most digital assets. Retain C for the intended unbacked-crypto context, not for all digital assets such as tokenized bonds.

Q52 分类确认 / Classification confirmed

不是需要改答案的错题。所附答案明确把改造、出租空置、重新定位的组合归为Value-add，保留B。

No answer change is needed. The supplied explanation explicitly classifies upgrades, vacancy leasing and repositioning as Value-add, so B is retained.

资料来源与完整性核对

Sources and completeness checks

主要来源是用户提供的两份PDF。每一道原题旁均标注题库页码与答案页码；单元知识总结用Q编号回指原题。以下补充资料仅支持勘误与适用语境提示。

The primary sources are the two user-supplied PDFs. Each question cites its question-bank and answer-file pages; concept notes refer back through Q numbers. The supplemental sources below support only correction and context notes.

文件 File	已使用范围 Material used
1000题(6).pdf Question source	总PDF104-114页，完整Q1-Q61及全部选项。 PDF pages 104-114, all stems and options for Q1-Q61.
另类投资(2).pdf Answer source	PDF1-61页，逐题答案及相关解析。 PDF pages 1-61, all answer keys and relevant explanations.

[S1] SEC / Investor.gov

Leveraged Loan Funds - Investor Bulletin | 2019-11-20

仅用于Q9术语校核。
Used only to check the terminology in Q9.
https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/investor-bulletins/leveraged-loan-funds-investor-bulletin

[S2] US Securities and Exchange Commission

Statement on the Approval of Spot Bitcoin Exchange-Traded Products | 2024-01-10

仅用于Q2原解析的时间语境说明。
Used only for the time-context note on Q2.
https://www.sec.gov/newsroom/speeches-statements/gensler-statement-spot-bitcoin-011023

[S3] Ethereum developer documentation

Smart contract security | 访问日期 / Accessed 2026-11-02

仅用于Q1智能合约安全性校核。
Used only to qualify the security claim in Q1.
https://ethereum.org/developers/docs/smart-contracts/security

完成核对 / Completion checks

61/61 原题完整，A/B/C选项齐全。
61/61 original questions have complete A/B/C options.

61/61 答案与所附答案文件匹配；题号与答案页码对应。
61/61 answers match the supplied key and page mapping.

61/61 均有中文题干、选项及双语解析与易错点。
61/61 have Chinese stems and options, bilingual explanations and traps.

6/6 费用计算独立重算；8个教学单元覆盖全部题目且不重复归类。
6/6 fee cases independently recalculated; eight units cover all questions with one primary placement each.

未加入自行编写或改编的习题；资料年份与出处不明之处已披露。
No invented or adapted questions are added; unknown source edition and provenance are disclosed.